HyperKING: Quantum-Classical Generative Adversarial Networks for Hyperspectral Image Restoration

Chia-Hsiang Lin, *Senior Member, IEEE*, and Si-Sheng Young, *Student Member, IEEE*

***Abstract*— Quantum machine intelligence starts showing its impact on satellite remote sensing (SRS). Also, recent literature exhibits that quantum generative intelligences encompass superior potential than their classical counterpart, motivating us to develop quantum generative adversarial networks (GANs) for SRS. However, existing quantum GANs are restricted by the limited quantum bit (qubit) resources of current quantum computers and process merely a small 2 × 2 grayscale image, far from being applicable to SRS. Recently, the novel concept of hybrid quantum-classical GAN, a quantum generator with a classical discriminator, has upgraded the order to 28 × 28 (still grayscale), whereas it is still insufficient for SRS. This motivates us to design a radically new hybrid framework, where both the generator and discriminator are hybrid architectures. We demonstrate this feasibility, leading to a breakthrough in processing 128 × 128 hyperspectral images (HSIs) for SRS. Specifically, we design the quantum part with mathematically provable quantum full expressibility (FE) to address core signal processing tasks, wherein the FE property allows the quantum network to realize any valid quantum operator with appropriate training. The classical part, composed of convolutional layers, treats the read-in (compressing the optical information into limited qubits) and read-out (addressing the quantum collapse (QC) effect) procedures. The proposed innovative hybrid quantum GAN, named "Hyperspectral Knot-like IntelligeNt dIscrimiNator and Generator" (HyperKING), where "knot" partly symbolizes the quantum entanglement and partly the compressed quantum domain in the central part of the network architecture. HyperKING significantly surpasses the classical approaches in hyperspectral tensor completion, mixed noise removal (about 3 dB improvement), and blind source separation (BSS) results.**

***Index Terms*— Generative adversarial network (GAN), hyperspectral image (HSI) restoration, mixed noise removal, quantum computing, quantum image processing, quantum machine learning.**

I. Introduction

QUANTUM remote sensing technology has been very recently developed for real-time hyperspectral satellite data restoration, where quantum computing has also been shown to be promising for developing high-performance satellite remote sensing (SRS) software [1]. To understand the strength of quantum computing, we remark that for a specific sampling task that would take about eight years in current computing facilities, the well-known "Zuchongzhi" quantum computer can accomplish the task in approximately 1 h [2]. Quantum computing is also effective in addressing NP-hard problems, such as automatic financial crime detection from the transaction records [3], fast searching via the well-known Grover's algorithm [4], as well as the factorization of an integer into prime factors using the Shor's algorithm [5]. Beyond these quantum applications, another leading approach is quantum machine intelligence, or more specifically, quantum deep network (QUEEN) [1], [6]. For instance, QUEENs significantly improve the performance of graph and convolutional neural networks (CNNs) by integrating the quantum unitary-computing and classical deep features in very recent SRS literature [7], [8]. Remarkably, QUEEN-inspired quantum prism has even been adopted to generate more virtual spectrum observations in order to solve an NP-hard underdetermined blind source separation (BSS) problem for optical satellite data analysis [9]. Another critical SRS application is object counting [10], whereas the relatively tiny shape of remotely sensed objects hinders effective and precise counting [11]. In the future, this challenging task may be effectively and precisely achieved by exploring other entanglement mechanisms for the design of QUEENs.

Given the successes of QUEENs, we aim to advance further by developing quantum-based generative artificial intelligence (AI). Specifically, among the various types of generative AIs (e.g., variational autoencoder or diffusion model), generative adversarial network (GAN) is regarded as the most prestigious and representative one with great successes over the past decade [12]. Unfortunately, traditional GANs often suffer from prohibitive training datasets and hardware requirements [13], leading to inefficient implementations. Recently, numerous studies [13], [14], [15] have demonstrated that quantum-based GANs encompass great potential to outperform their classical counterparts. This fact will also be qualitatively and quantitatively verified on the challenging mixed noise removal tasks (see Section III-F). Consequently, constructing the quantum-based GANs has gained considerable attention and become an emerging topic in the SRS field.

Received 18 December 2024; revised 18 March 2025; accepted 14 April 2025. Date of publication 17 April 2025; date of current version 9 May 2025. This work was supported by the Emerging Young Scholar Program (namely, the 2030 Cross-Generation Young Scholars Program) of National Science and Technology Council (NSTC), Taiwan, under Grant NSTC 113-2628-E-006-003. *(Corresponding author: Chia-Hsiang Lin.)*

Chia-Hsiang Lin is with the Department of Electrical Engineering and the Miin Wu School of Computing, National Cheng Kung University, Tainan 70101, Taiwan (e-mail: chiahsiang.steven.lin@gmail.com).

Si-Sheng Young is with the Institute of Computer and Communication Engineering, Department of Electrical Engineering, National Cheng Kung University, Tainan 70101, Taiwan (e-mail: q38121509@gs.ncku.edu.tw).

Digital Object Identifier 10.1109/TGRS.2025.3561951

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON AMONG THE PROPOSED HYPERKING AND EXISTING QUANTUM-BASED GANS FOR IMAGE PROCESSING

Methods	Dataset	Image Size	Image Type	Compression	Hybrid State (Generator / Discriminator)
QuGAN [16]	MNIST Image	2×2	Grayscale (single band)	PCA Feature	Quantum / Quantum (Non-Hybrid)
Patch GAN [13]	Handwritten Digits Image	8×8	Grayscale (single band)	Patch-and-Batch	Quantum / Classical (Hybrid)
PQWGAN [15]	Fashion MNIST Image	28×28	Grayscale (single band)	Patch-and-Batch	Quantum / Classical (Hybrid)
HyperKING	NASA's EO-1 Satellite Image	$128 \times 128 \times 172$	Hyperspectral (>100 bands)	Deep Feature	(Quantum-Classical) / (Quantum-Classical) (Hybrid)

To further clarify our research aim, we first remark that quantum GAN was conceptually proved to be a feasible approach, though still far from being applicable to the processing of large images. Typically, a quantum GAN comprises two players, i.e., quantum generator and quantum discriminator. The former aims to produce plausible data, while the latter tries to distinguish the statistical discrepancy between the real data and the fake one generated from the generator (see Fig. 1). Under the quantum adversarial learning (QAL), the dynamic game would converge to some Nash equilibrium yielding a capable quantum generator with quite accurate distributions. Nevertheless, the existing quantum GAN could only process small-sized grayscale images due to the inevitable restrictions of near-term quantum computers, and is hence not applicable to the multichannel image processing in SRS.

To better understand the challenge, we summarize the comparison among the existing quantum-based GANs in Table I. For instance, the pioneering QuGAN [16] returns only 2×2 samples per run, followed by performing inverse principal component analysis (PCA) to complete the image generation. Subsequently, by leveraging the innovative concept of hybrid quantum-classical GAN (i.e., quantum generator with classical discriminator), the order of resolution is directly upgraded to 8×8 grayscale pixels [13]. Later on, the hybrid GAN further achieves a resolution of 28×28 for grayscale image generation, as proposed recently in [15]. These all demonstrate the effectiveness of the hybrid approach, most importantly emphasizing the necessity of complementary techniques of quantum computing, which enables advanced functionalities on near-term quantum computers, as alluded in [17]. For example, deep learning has served as a complementary technique to assist quantum computing in achieving complicated quantum image processing tasks like salt-and-pepper noise removal [1].

However, these hybrid GANs still fall within the scope of grayscale image processing [15], marking a deficient spectral capacity of the existing quantum-based GANs. Whereas optical SRS, in particular, hyperspectral SRS, highly relies on the spectral capacity for accurate object identification [18]. Before we further specify the challenges of trying to apply these approaches for SRS, we would like to ask a simple question: *If the hybrid architecture is so effective, why not just bring the hybrid design directly into the generator (and also the discriminator)?* This motivates us to design a radically new hybrid GAN architecture, directly making both generator and discriminator *hybrid*. In other words, existing methods all have the hybrid form with respect to (w.r.t.) the entire GAN, while each network (generator or discriminator) still remains either quantum or classical architecture, as summarized in Table I. We aim to demonstrate that our new design philosophy is feasible, and our proposed quantum-classical GAN leads to a breakthrough of not only achieving 128×128 spatial resolution but also up to 172-channel hyperspectral SRS image processing (much upgraded compared to the single-channel grayscale image).

Before we elaborate on the design of the hybrid GAN, we briefly recall the challenges of current quantum computers to clarify the practicality of the proposed approach. First, the limited qubit resources prevent the near-term quantum computers from well processing large images. In fact, the number of entangled qubits could be quite limited to be within 100 for current quantum computers [19]. Specifically, the well-known "IBM Eagle" is composed of just 127 qubits, while IBM has announced another quantum computer "IBM Osprey" with 433 qubits. The key scientific issue here is how to build quantum mechanics allowing multiple particles to stay entangled. On the other hand, when one attempts to read out the quantum image state from the quantum computer, it will collapse to some eigenstate, making the original quantum state not directly observable. Under these practical constraints, this work aims to demonstrate that hundred-scale qubits would be sufficient to realize the quantum generative AI with advanced hyperspectral SRS applications.

To design our hybrid quantum-classical generator, we employ the low-rank nature of hyperspectral data to design a specific QUEEN, leading to an architecture that first processes the spatial information and then the spectral information. Specifically, the classical part of the proposed generator plays a role in compressing the signal into the highly compressed quantum feature space, making the quantum image processing feasible under the limited qubit resources. In comparison, the PCA compression used in existing quantum GANs may only preserve 28% data variance for the QUEEN to generate images [16]. If we adopt this less effective compression (i.e., dividing SRS images into several grayscale patches) for SRS applications, the generated images would not preserve high-fidelity spectral characteristics, potentially leading to serious misidentification for some spectrum-driven SRS tasks, such as hyperspectral anomaly detection [20]. Other quantum-based GANs mostly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

using patchwise processing strategy (see Table I) overlook the importance of spatial relations in processing SRS images. Such a limitation may hinder those critical SRS tasks that highly rely on spatial continuity. For instance, in hyperspectral change detection, the changes are known to be coherently associated with their neighboring regions [21]. Different from the abovementioned compression approaches for addressing the limited qubit resources, we adopt the classical components to effectively and efficiently compress the spectral and spatial information simultaneously; subsequently, the quantum part of our hybrid generator performs the core quantum signal processing tasks. Remarkably, unlike the conventional deep learning module that is often considered as a blackbox with unknown capability, our hybrid generator cleverly deploys quantum neurons on the deep network to ensure the quantum full expressibility (FE) of the proposed core quantum module. The FE property means that for any valid quantum unitary operator, there exist network parameters such that the core quantum module exactly implements the operator. The related theoretical guarantee of FE will be detailed in Section II. After the core quantum signal processing stage, the classical deep learning is adopted to learn the inverse mapping from the collapsed eigenstate back to the target quantum image state, thereby mitigating the quantum collapse (QC) effect and again emphasizing the importance of introducing the complementary techniques for implementing advanced tasks on near-term quantum computers. Overall, our hybrid generator has a knot-like shape, where the term "knot" partly symbolizes the concept of quantum entanglement and partly symbolizes the highly compressed quantum feature space (wherein core quantum signal processing happens) for saving the qubit resources.

As for the design of our hybrid classifier, it actually relates to another challenging issue, which is to ensure that the proposed hybrid generator and discriminator can help each other to alternatively evolve to stronger statuses under the QAL. To this end, the hybrid discriminator may be simply considered and constructed as a highly entangled (HE) deep classifier. Specifically, the function of the quantum-driven discriminator is regarded as to classify a hyperspectral image (HSI) to be fake or real (see Fig. 1), and this function should be sufficiently strong for the discriminator to adversarially help the proposed hybrid generator to be trained stronger under the dynamic game theory framework [22]. The proposed hybrid discriminator is composed of a series of quantum rotation gates in an HE form (just like a knot), which is known to be an effective strategy for the classification task [23]. The key challenge here is to ensure that our discriminator is neither too strong nor too weak, so that the generator and discriminator can be successfully evolved via the QAL game [24]. Comprehensive analyses, as summarized in Section III-B, prove the balanced competitiveness between the proposed generator and discriminator. Once the game converges, the hybrid generator is expected to be strong enough to be applicable to challenging tasks (e.g., hyperspectral satellite image restoration). The proposed method is termed "Hyperspectral Knot-like IntelligeNt dIscrimiNator and Generator" (HyperKING), which is the first quantum generative AI that achieves advanced non-grayscale

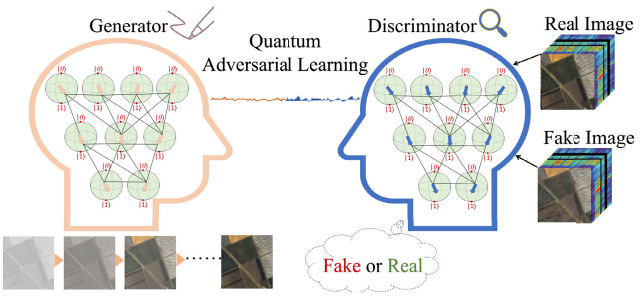


Fig. 1. Illustration of the QAL via game theory, where the two players are the quantum generator and the quantum discriminator. The former aims at generating something plausible, while the latter should judge whether something is real of fake. Both the generator and discriminator have quantum logical gates as their neurons.

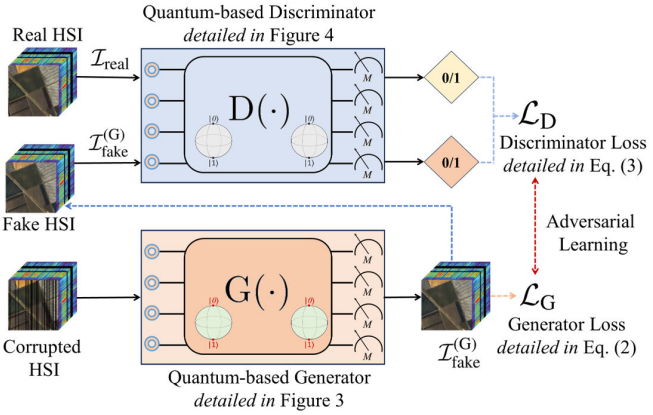


Fig. 2. Overall framework of the proposed HyperKING, where G(·) and D(·) denote the function of the quantum-based generator and discriminator, respectively. QAL is conducted through the adversarial loss functions between the two competitive players.

quantum image processing tasks. To better understand our advanced quantum-based generative AI, the overall architecture of HyperKING is provided in Fig. 2.

The remaining parts of this article are organized as follows. In Section II, we propose a novel quantum-classical generative AI theory and framework, including the design of the hybrid generator, the design of hybrid discriminator, and the mathematical guarantee of quantum expressibility with mathematical details collectively provided in Section II-B. In Section III, we apply our theory to show that quantum-driven SRS is feasible, demonstrate that the proposed hybrid generator and discriminator do have competitive ability to help each other to evolve to a stronger state, and prove that the trained hybrid generator achieves good results in hyperspectral tensor completion, mixed noise removal, and BSS. Concluding remarks are drawn in Section IV.

II. Proposed Quantum-Based Generative AI Framework

Quantum-based machine intelligence starts showing its impact on optical SRS, as featured by the recent achievement of the high-accuracy QUEEN-based satellite data restoration. This section aims to go a step further to develop quantum-based generative AI with sufficiently large spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

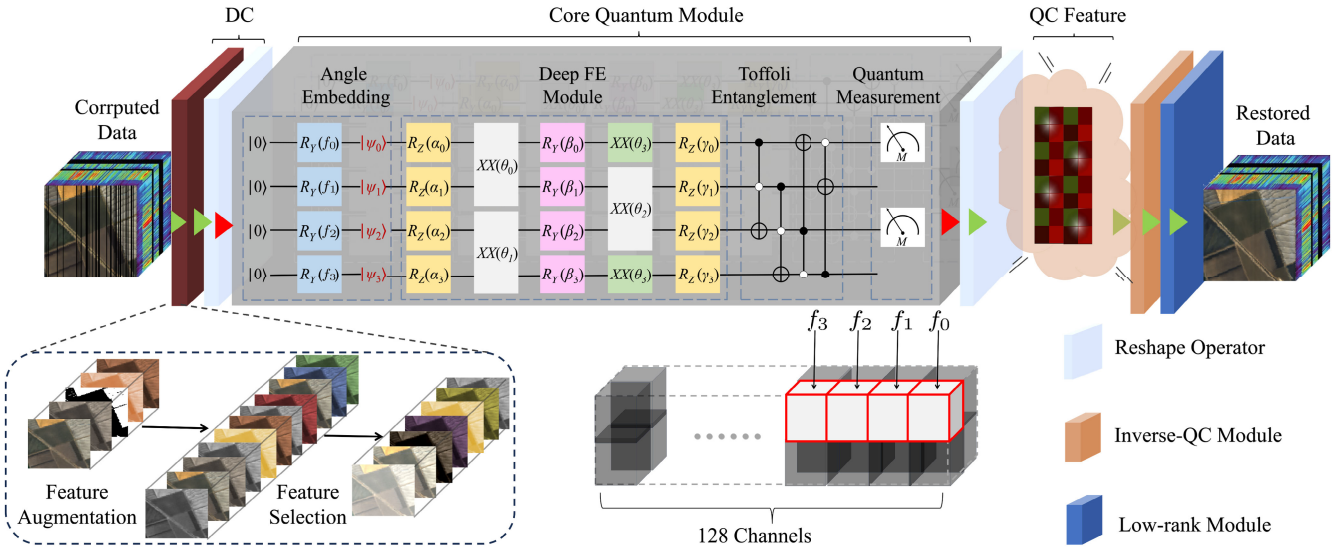


Fig. 3. Proposed hybrid generator for HSI processing, where the QC effect and the issue of limited qubit resource are well addressed as experimentally demonstrated. The FE of the Ising-Rotation architecture "$R_Z - XX - R_Y - XX - R_Z$" is theoretically proved. Please refer to [1, Fig. 3].

TABLE II

FREQUENTLY USED QUANTUM GATES, THEIR SYMBOLS AND THE CORRESPONDING UNITARY OPERATORS [25], [26], WHERE $c_\theta \triangleq \cos(\theta/2)$, $s_\theta \triangleq \sin(\theta/2)$, AND $\mathrm{DIAG}(\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_N)$ DENOTES THE BLOCK-DIAGONAL MATRIX WITH $\boldsymbol{Y}_n$ BEING THE nTH DIAGONAL BLOCK FOR $n = 1, \ldots, N$ [27]

Quantum Gate	Symbol	Unitary Operator
Rotation X	$R_X(\theta)$	$\begin{pmatrix} c_\theta & -is_\theta \\ -is_\theta & c_\theta \end{pmatrix}$
Rotation Y	$R_Y(\theta)$	$\begin{pmatrix} c_\theta & -s_\theta \\ s_\theta & c_\theta \end{pmatrix}$
Rotation Z	$R_Z(\theta)$	$\begin{pmatrix} e^{-i(\theta/2)} & 0 \\ 0 & e^{i(\theta/2)} \end{pmatrix}$
Ising XX	$XX(\theta)$	$\begin{pmatrix} c_\theta & 0 & 0 & \frac{s_\theta}{i} \\ 0 & c_\theta & \frac{s_\theta}{i} & 0 \\ 0 & \frac{s_\theta}{i} & c_\theta & 0 \\ \frac{s_\theta}{i} & 0 & 0 & c_\theta \end{pmatrix}$
Pauli-X	X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Pauli-Z	Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Toffoli		$\mathrm{DIAG}(\boldsymbol{I}_4, X, \boldsymbol{I}_2)$
CRX	R_X	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & c_\theta & -is_\theta \\ 0 & 0 & -is_\theta & c_\theta \end{pmatrix}$

capacity, in order to perform meaningful texture analysis in advanced image processing tasks. Existing quantum GAN could only process a small 2×2 grayscale image or single-channel image; after introducing the novel concept of hybrid quantum-classical GAN (see Table I), the resolution is upgraded to 8×8 and very recently advanced to 28×28 grayscale pixels. This motivates us to consider the hybrid structure but using a radically new hybrid design, directly making both generator and discriminator in the hybrid form (see Table I). Comprehensive experiments, as summarized in Section III, will demonstrate that our new design does lead to a breakthrough in processing 128×128 SRS image with 172-channel hyperspectral bands.

In our proposed quantum-based generative AI, i.e., HyperKING, the quantum part is responsible for the core signal processing tasks with mathematically provable quantum FE, while the classical part treats the read-in (compressing the optical information into limited qubits) and read-out (addressing the QC effect) procedures, leading to the "knot-like" architecture as discussed in Section I. In HyperKING, both the highly compressed central quantum part of the generator (see Fig. 3) and the HE structure of the discriminator (see Fig. 4) are symbolized by the term "knot."

To understand the key principles behind our design, preliminary knowledge of quantum computing and quantum deep learning is required for a comprehensive understanding. We refer interested readers to [1, Section II.B] and [1, Section II.C], where one can find a concise review of preliminary backgrounds, including the quantum Dirac notation system, quantum measurement, QC effect, barren plateaus effect, Bloch sphere, and quantum deep learning, etc. Sections II-A and II-B describe the design philosophy of the proposed quantum-based generative AI, detailing the proposed hybrid generator and hybrid discriminator, followed by some analysis and discussion. By illustrating our design philosophy, we aim to shed some insights for future investigators in this pioneering research line.

A. HyperKING: The Design of Hybrid Generator and Hybrid Discriminator

Generative AI has found numerous applications in various domains [28]. Oriented from the game theory, a game is composed of three components—the players in the game, their

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

strategy spaces, and the utility functions [22]. As a typical example of generative AI, there are two players (or networks), which are referred to as the generator and the discriminator. The strategy space of the generator is the parameter space of the generative network, aiming at generating plausible samples that are capable of misleading the discriminator. The strategy space of the discriminator is the parameter space of the discriminative network, aiming at discriminating whether given samples are real or fake. Eventually, the utility of the generator is evaluated by its capability of deceiving, while the utility of the discriminator is evaluated by its resilience against being deceived. The above idea is graphically illustrated in Fig. 1. As for the proposed quantum-based generative AI, quantum neurons (i.e., quantum gates summarized in Table II) are used to deploy the hybrid generator/discriminator networks. To the best of our knowledge, the demonstration of quantum-based generative AI for processing multichannel, sufficiently large images (needed in real-world SRS applications) is absent. For the first time, we design an advanced hybrid GAN framework and apply it to advanced tasks in SRS.

To design the hybrid generator, recall that our target application is to restore hyperspectral data, meaning that the generator needs to output an HSI that is, however, known to be of low rank. Under the convex optimization framework, such a low rankness can be forced by suitable regularization functions like nuclear norm [27]. However, enforcing a generative network to return a low-rank HSI, denoted as $\boldsymbol{X}$, is relatively challenging. This is equivalent to requiring the network itself to fulfill the mission of low-rank regularization, which is closely related to the well-known concept called deep image prior (DIP) [29]. Instead of designing some convex regularization/loss functions [30], the DIP concept argues that the network itself actually implements some specific regularization. Inspired by the low-rank modeling $\boldsymbol{X} = \boldsymbol{AS}$ [31] wherein $\boldsymbol{A}$ encodes the spectral information, while $\boldsymbol{S}$ encodes the spatial details, our hybrid generator first addresses the spatial details $\boldsymbol{S}$ through a customized QUEEN, and then performs the spectral upsampling (namely, left-multiplying by $\boldsymbol{A}$) at the last classical layer as illustrated in Fig. 3 and detailed in Table III, i.e., low-rank module E_r [1].

Besides the low-rank design, deep learning also offers a great environment for saving the limited qubit resources. Specifically, deep learning can be regarded as to process the signal in the highly compressed feature space, rather than directly processing the original hyperspectral signal of huge data volume. This idea of deep compression (DC) is more effective than patchwise computing or PCA compression used in existing benchmark quantum-based GANs (see Table I), and is expected to capture more nonlinear effects of real-world satellite data for subsequent quantum processing.

Thus, at the beginning of the proposed hybrid generator, we add a DC module to transform the hyperspectral information into the highly compressed feature space, wherein we conduct the core quantum information processing (i.e., the core quantum module), as illustrated in Fig. 3 and detailed in Table III, where "MCB(x, y)" denotes the multiple convolution block (CB) composed of 2×2 max pooling, CB(x,3,0), CB($2x$,3,1), and CB($2x$,3,0)$\times y$; note that CB(c, s, p) is defined in Table III. Before formally designing the core quantum module, we first discuss the QC effect, which prevents us from smoothly reading out the well processed/restored quantum image from the quantum computer. To tackle this challenge, we adopt the inverse-QC module to learn the inverse mapping from the measured QC features back to the original data space, as illustrated in Fig. 3 and detailed in Table III. With all the above keeping in mind, we are ready to design the quantum components (i.e., core quantum module), the most challenging and tricky part for the core quantum signal processing tasks.

First of all, the core quantum module adopts the quantum angle encoding scheme [32], for which each feature element f is encoded as the angle of the quantum rotation gate R_Y (i.e., $R_Y(\theta)|_{\theta:=f}$) for the subsequent quantum information processing (see Fig. 3), where R_Y is defined in Table II. The way we treat the quantum features is graphically illustrated in Fig. 3. One can see that we are not treating one channel at once, but treating f_k's from different channels at once. The reason is that the success of classical deep learning would come from the interaction among different feature maps (i.e., channels), allowing the extraction of more abstract features in the subsequent layers (i.e., deeper layers), and our way of treating the quantum feature allows more interaction among different channels. Second, we deploy the five-layer "$R_Z - XX - R_Y - XX - R_Z$" architecture to process the quantum features. Unlike most deep learning works wherein the network expressibility is often unknown (blackbox), we prove that the above Ising-Rotation architecture has FE, meaning that the deployed quantum neurons are able to express/implement any valid quantum unitary operator. This is rigorously stated in the following theorem, whose proof is relegated to Section II-B.

Theorem 1: The learnable quantum neurons deployed in the core quantum module of the proposed hybrid generator (see Fig. 3) can realize any valid quantum operator U, with some real-valued network parameters $\{\alpha_k, \beta_k, \gamma_k, \theta_k\}$. □

Third, the core quantum module adopts the Toffoli entanglement, implemented by the controlled CNOT (CCNOT) with the controlled state $|10\rangle$, followed by the Pauli-Z quantum measurement (see Table II) [1]. Finally, to see the insight from Theorem 1, we remark that from an intuitive ground, being able to express comprehensive quantum operators (i.e., FE) has a positive impact on "generating" more plausible images—a desired property for GAN.

As for the design of our hybrid discriminator, we strategically reformulate its function as a classifier that determines whether a given HSI is real or generated. This allows us to judiciously design the discriminator as a quantum-based classifier. The most challenging part would be how to ensure that the designed discriminator has a comparable ability to the proposed hybrid generator; otherwise, the adversarial learning cannot be effectively proceeded, making the trained generator not strong enough to tackle the challenging satellite data restoration mission in SRS. For this aim, we propose the following architecture for our hybrid discriminator, which is graphically illustrated in Fig. 4.

First, before the quantum amplitude encoding [33], we propose the downsampling (DS) module that takes merit of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

max pooling to have a more concise image representation for saving the qubit resources, while keeping the most significant local information. However, considering that accurate classification would rely on some details removed by the max pooling, we mitigate the concern by further keeping those information via adding the average pooling, as displayed in Fig. 4 and detailed in Table IV. Second, the core of the hybrid discriminator is designed as an HE quantum classifier. To this end, we define the entanglement module EM($bcd \mid a$) that uses qubit a to sequentially control the other qubits (b, c, d) through the controlled rotation X (CRX) quantum gates (see Fig. 4), where CRX has been defined in Table II. We use consecutive EM modules to achieve the desired HE property, and this HE strategy is known to be effective as investigated in [23]. Third, the quantum measurement is conducted by the Pauli-X quantum gates, followed by the sigmoid layer that transforms measured quantum feature maps into probability maps, from which the hybrid discriminator determines whether the input HSI is real or generated. The detailed structure of the proposed hybrid discriminator is given in Table IV. Comprehensive experiments, as summarized in Section III-B, will evaluate the competitiveness of this elegant design compared to our hybrid generator, demonstrating the effectiveness in training the hybrid discriminator and hybrid generator through the QAL.

Our experiments also provide an insight for the design of QAL because the capability of our hybrid generator (trained with the QAL) in various SRS restoration tasks is never a coincidence. The experiments in Section III will show that when the hybrid discriminator is revised to have more sophisticated or more simplified quantum structures, the QAL may not be smoothly done. Thus, striking a balance between the generator and discriminator is one of our key contributions, allowing the two players to dynamically help each other to get stronger in the quantum adversarial game.

We call our quantum generative AI framework as "Hyperspectral Knot-like IntelligeNt dIscrimiNator and Generator" (HyperKING)! The term "knot" is partly to symbolize quantum stuffs that are entangled together (e.g., the Toffoli-entangled qubits in the generator, or the CRX-entangled qubits in the discriminator), and is partly to use the compressed central part of a knot (or bowknot) to symbolize the proposed highly compressed quantum feature space for saving the qubit resources. HyperKING is implemented on Pytorch and PennyLane platforms (the leading tool for programming quantum computers) [34], [35], and is trained using the well-known smoothed ℓ_1-norm-aided entropy loss function [36] (defined in Section III) with the root mean square propagation (RMSProp) optimizer for addressing the non-stationary adversarial learning [37]. The superiority of the proposed HyperKING will be demonstrated in Section III.

TABLE III

ARCHITECTURE OF THE PROPOSED HYBRID GENERATOR. "CB(c, s, p)" DENOTES CB COMPOSED OF THE CONVOLUTION LAYER "CONV(c, s, p)," BATCH NORMALIZATION (BN), AND LEAKYRELU (WITH NEGATIVE SLOPE 0.2), WHILE "TCB(c, s)" DENOTES TRANSPOSED CB COMPOSED OF TRANSPOSED CONVOLUTION TCONV(c, s), BN, AND LEAKYRELU, WHERE (c, s, p) SPECIFIES THE NUMBER OF OUTPUT CHANNELS, THE KERNEL SIZE OF $s \times s$, AND THE NUMBER OF PADDINGS, RESPECTIVELY. THE UPSAMPLING LAYER IS IMPLEMENTED BY BILINEAR INTERPOLATION

	Layer	Configuration	Output Size
	Input	-	172×128×128
DC Module	ConvModule 1	CB(516,3,1)×1 CB(172,3,1)×1 CB(32,3,1)×1 CB(16,3,0)×2	16×124×124
	ConvModule 2	MCB(16,2)×1	32×56×56
	ConvModule 3	MCB(32,3)×1	64×20×20
	ConvModule 4	MCB(64,3)×1	128×2×2
Core Quantum FE Module (Theorem 1)	Reshape	-	128×4
	Data Embedding	$R_Y(128, 4)$	128×4
	Unitary Gate 1	$R_Z(128, 4)$	128×4
	Reshape	-	256×2
	Unitary Gate 2	$XX(256, 2)$	256×2
	Reshape	-	128×4
	Unitary Gate 3	$R_Y(128, 4)$	128×4
	Reshape	-	256×2
	Unitary Gate 4	$XX(256, 2)$	256×2
	Reshape	-	128×4
	Unitary Gate 5	$R_Z(128, 4)$	128×4
	Toffoli Entanglement	CCNOT(0,1,2)	128×4
		CCNOT(1,2,3)	128×4
		CCNOT(2,3,0)	128×4
		CCNOT(3,0,1)	128×4
	QC Measurement	$Z(128, 2)$	128×2
	Reshape	-	64×2×2
Inverse-QC Module	TConvModule 1	TCB(64,3)×4 2×2 Up-sample	64×20×20
	TConvModule 2	TCB(64,3)×2 TCB(32,3)×2 2×2 Up-sample	32×56×56
	TConvModule 3	TCB(32,3)×2 TCB(16,3)×1 2×2 Up-sample	16×124×124
	TConvModule 4	TCB(16,3)×1 TCB(8,3)×1	8×128×128
Low-rank Module	E_r Module	CB(172,1,0)×1 Conv(172,1,0)×1	172×128×128

B. Proof of Theorem 1

Before we formally detail the proof of Theorem 1, we provide an intuitive explanation for each step to enhance the understanding of readers unfamiliar with quantum computing. In *Step 1*, based on the basic properties of quantum computing summarized in [1, Section II.A] and [1, Section II.B], we first define the general representation of the core quantum module U with specific real-valued network parameters $\{\alpha_k, \beta_k, \gamma_k, \theta_k\}$. Then, we only need to show that for any unitary matrix V (representing a specific valid quantum function), it can be realized by the core quantum module U through adapting its parameters, or more precisely, training its parameters. To this end, we require the key properties established in *Step 2*. The first property, as formulated in (1), demonstrates that V can be decomposed into the $R_Z - R_Y - R_Z$ architecture (i.e., FE property) without involving quantum

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

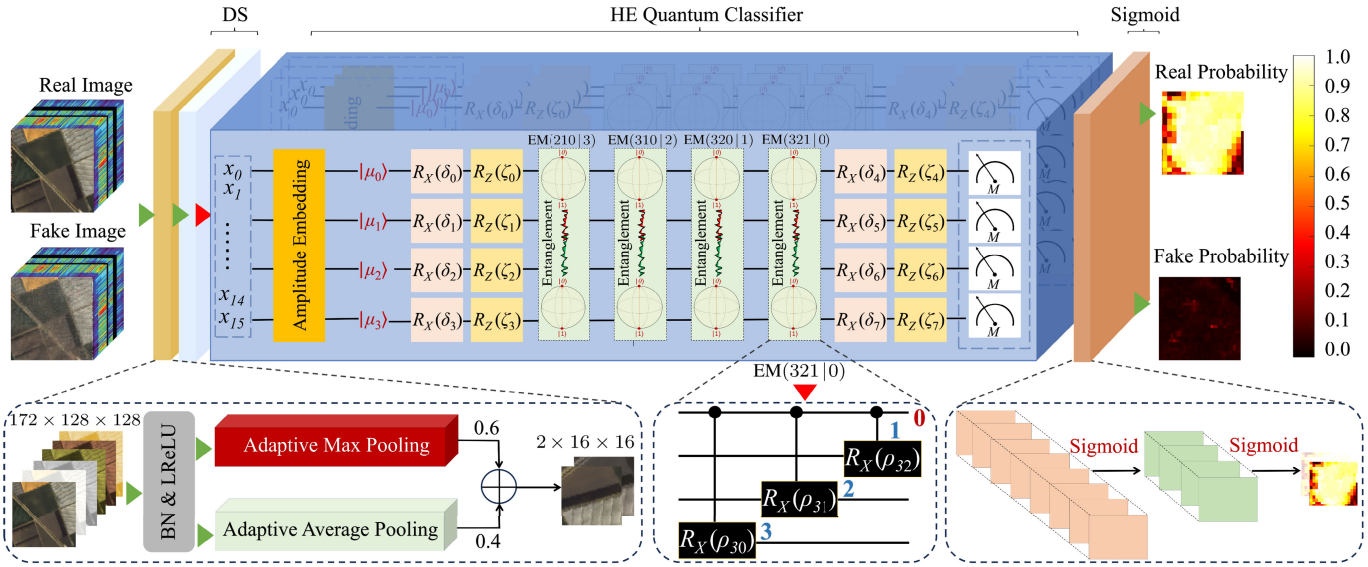


Fig. 4. Proposed hybrid discriminator for HE quantum classification, where the HE property is achieved by the entanglement module EM($bcd\,|\,a$) meaning that qubit a controls qubits (b, c, d) via the controlled rotation X (CRX) gates.

TABLE IV

CONFIGURATION OF OUR HYBRID DISCRIMINATOR (SEE FIG. 4). "LRELU(m)" DENOTES LEAKYRELU WITH NEGATIVE SLOPE m. "APOOL(x, y, z)" DENOTES ADAPTIVE POOLING MODULE COMPOSED OF ADAPTIVE MAX POOLING "ADAPTIVEMAXPOOL(x, y, z)" AND ADAPTIVE AVERAGE POOLING "ADAPTIVEAVGPOOL(x, y, z)," OUTPUTTING A x-CHANNEL TENSOR OF SPATIAL SIZE $y \times z$. QUANTUM AMPLITUDE EMBEDDING "AMP(x, y)" ENCODES x FEATURES OF DIMENSION y INTO x QUBIT STATES OF DIMENSION $\log_2(y)$. THE MODULE "SIGM(m, n)" IS COMPOSED OF A TRAINABLE $m \times n$ MATRIX FOR LINEAR PROJECTION, FOLLOWED BY A SIGMOID FUNCTION

	Layer	Configuration	Output Size
	Input	-	172×128×128
DS Module	Pooling Layer	BN×1 LReLU(0.2)×1 APool(2,16,16)×1	2×16×16
HE Quantum Classifier Module	Reshape	-	32×16
	Data Embedding	Amp(32,16)	32×4
	Unitary Gate 1	$R_X(32, 4)$	32×4
	Unitary Gate 2	$R_Z(32, 4)$	32×4
	CRX Entanglement	EM(210|3)	32×4
		EM(310|2)	32×4
		EM(320|1)	32×4
		EM(321|0)	32×4
	Unitary Gate 3	$R_X(32, 4)$	32×4
	Unitary Gate 4	$R_Z(32, 4)$	32×4
	QC Measurement	$X(32, 4)$	32×4
	Reshape	-	1×128
Sigmoid Module	Sigmoid Layer	Sigm(128,16)×1 Sigm(16,1)×1	1×1

entanglements under a single-qubit case. However, to accomplish the challenging SRS restoration task, the network may require powerful quantum entanglements to facilitate a specific task. To tackle it, we further deploy the trainable entanglement gate, Ising XX, in the core quantum module, where Ising gate may be considered as a soft entanglement not really forcing the entanglement but forcing cross-qubit feature interactions. Specifically, as to be presented in *Step 2*, the Ising gate could occasionally become transparent as $\theta_k := 0$, for which the quantum module reduces back to the above naive FE form. This implies that the network can adaptively conduct the entanglement-like interactions as needed to achieve effective restoration. In *Step 3*, we extend the single-qubit case to the multiple-qubit case by elegantly adopting some properties of the Kronecker product, and hence completing the proof of Theorem 1. With the above intuition and outline in mind, we are ready to mathematically describe the proof of Theorem 1 in the following steps.

Step 1: All the quantum gates used below have been defined in Table II. To facilitate the proof, we further define the notation $XX_{mn}(\theta)$, which means that the Ising XX gate $XX(\theta)$ is operated on the mth and nth qubits. Accordingly, let $I_0(\theta_2, \theta_3) \triangleq XX_{12}(\theta_2) \otimes XX_{03}(\theta_3)$ and $I_1(\theta_0, \theta_1) \triangleq XX_{01}(\theta_0) \otimes XX_{23}(\theta_1)$ represent the overall quantum effects of the two Ising layers in our hybrid generator. Similarly, let $A \triangleq \bigotimes_{k=0}^{3} e^{ip_k} R_Z(\alpha_k)$, $B \triangleq \bigotimes_{k=0}^{3} R_Y(\beta_k)$ and $C \triangleq \bigotimes_{k=0}^{3} R_Z(\gamma_k)$ represent the overall quantum effects of the three rotation layers in our hybrid generator. Furthermore, any valid quantum operator must be unitary (see Table II), because a quantum gate is built from certain Hamiltonian for a specific time, yielding a unitary time evolution based on the Schrödinger equation [26]. Thus, by letting U_k be any unitary operator acting on the kth qubit for $k \in \{0, 1, 2, 3\}$, it suffices to show that there exist real-valued network parameters $\{\alpha_k, \beta_k, \gamma_k, \theta_k\}$ such that

$$U \triangleq \bigotimes_{k=0}^{3} U_k = A I_0(\theta_2, \theta_3) B I_1(\theta_0, \theta_1) C$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

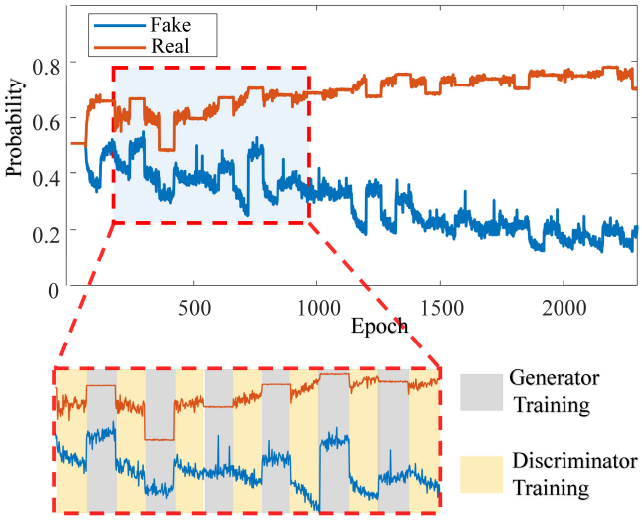


Fig. 5. For a given hybrid discriminator at a specific time/epoch, the probability for the discriminator to judge a real image (resp., fake image) to be real is displayed as the orange curve (resp., blue curve), and the results are averaged over 480 independent runs. The gray and yellow blocks represent the training periods for the hybrid generator and discriminator, respectively.

and this will be proven true (thereby completing the proof of Theorem 1) in the following steps.

Step 2: For any unitary matrix V, there exist real-valued phase p and parameters α, β, γ, such that

$$V = e^{ip} R_Z(\alpha) R_Y(\beta) R_Z(\gamma) \tag{1}$$

as stated and proved in [26, Th. 4.1]. Moreover, by the definition of Ising XX in Table II and the definition of $XX_{mn}(\theta)$ in Step 1, the setting of the real-valued parameter $\theta_k := 0$ yields

$$I_0(\theta_2, \theta_3) = XX_{12}(0) \otimes XX_{03}(0) = (\boldsymbol{I}_2 \otimes \boldsymbol{I}_2) \otimes (\boldsymbol{I}_2 \otimes \boldsymbol{I}_2)$$
$$I_1(\theta_0, \theta_1) = XX_{01}(0) \otimes XX_{23}(0) = (\boldsymbol{I}_2 \otimes \boldsymbol{I}_2) \otimes (\boldsymbol{I}_2 \otimes \boldsymbol{I}_2)$$

implying that we just need to show that there exist real-valued $(\alpha_k, \beta_k, \gamma_k)$ such that $U = ABC$ (cf. Step 1).

Step 3: According to (1), we can find real-valued $(\alpha_k, \beta_k, \gamma_k)$ to achieve the following unitary matrix decomposition, i.e.,

$$U_k = e^{ip_k} R_Z(\alpha_k) R_Y(\beta_k) R_Z(\gamma_k), \quad k = 0, \ldots, 3.$$

Then, we recall that for matrices $\boldsymbol{M}_1, \ldots, \boldsymbol{M}_4$ of proper dimensionality, the Kronecker product satisfies $(\boldsymbol{M}_1 \otimes \boldsymbol{M}_2)(\boldsymbol{M}_3 \otimes \boldsymbol{M}_4) = (\boldsymbol{M}_1 \boldsymbol{M}_3) \otimes (\boldsymbol{M}_2 \boldsymbol{M}_4)$. This property of Kronecker matrix equation, together with the matrices (A, B, C) defined in Step 1, gives the key relation of

$$ABC = \bigotimes_{k=0}^{3} \left[e^{ip_k} R_Z(\alpha_k) R_Y(\beta_k) R_Z(\gamma_k) \right]$$

or, equivalently, $U := \bigotimes_{k=0}^{3} U_k = ABC$, which completes the proof of Theorem 1. ■

III. Experimental Results

In this section, we comprehensively evaluate the performance of the proposed HyperKING quantum-based generative AI in the advanced SRS task (i.e., 172-channel HSI restoration). Before we introduce the baseline methods, we remark that existing quantum-based GANs still fall within the scope of grayscale image processing (single-channel), as discussed in Section I. Hence, they are not considered as baselines. Though the hyperspectral-driven QUEEN (HyperQUEEN) [1] is not a quantum-based GAN (not a quantum generative AI), HyperQUEEN does achieve advanced quantum image processing (AQIP). So, we employ the pretrained HyperQUEEN as a quantum-based baseline method. Beyond the quantum-based architecture, we also leverage the classical baselines, as there are rather limited AQIP algorithms. Some of them are optimization-based methods, including the low-rank tensor decomposition with total variation (LRTDTV) [38] and high-order tensor completion based on nuclear-norm and fast Fourier transform (HTNN-FFT) [39]. On the other hand, we also employ two DIP-based deep learning methods, i.e., hyperspectral DIP (HDIP) [40] and robust deep low-rank hyperspectral inpainting (R-DLRHyIn) [41].

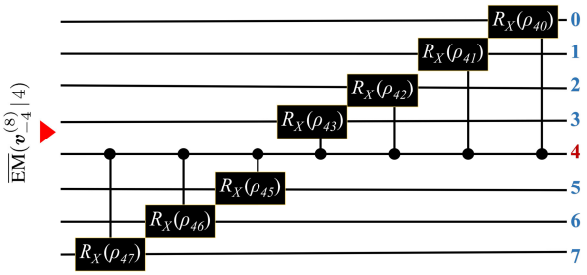


Fig. 6. Schematic of $\overline{\mathrm{EM}}(\boldsymbol{v}_{-4}^{(8)} \mid 4)$ embedded within sophisticated hybrid discriminator.

The remaining parts of this section are organized as follows. The description of experimental settings, including the brief introduction to the baselines, and detailed description of the training/testing data, are presented in Section III-A. Before we investigate the performance of the proposed HyperKING, an elaborate analysis of the comparability between the hybrid generator and the hybrid discriminator is presented in Section III-B. In Sections III-C–III-E, we perform the qualitative evaluation, quantitative evaluation, and BSS [i.e., hyperspectral unmixing (HU)] evaluation, respectively, to investigate the strength of the proposed HyperKING. Finally, we conduct hyperspectral denoising evaluation in Section III-F to further substantiate the effectiveness of our HyperKING.

A. Experimental Settings

Let us concisely introduce the benchmarks hereinafter. The pioneering AQIP architecture, i.e., HyperQUEEN, is the first quantum framework that solves the image restoration problem [1], but there is no quantum-driven discriminator to force HyperQUEEN to surge to a new level of AQIP performance. Another distinction between HyperKING and HyperQUEEN is the architecture of their core quantum module. Specifically, the generator in HyperKING is developed by the "$R_Z - XX - R_Y - XX - R_Z$" architecture (see Fig. 3), whereas HyperQUEEN comprises a "$R_Y - XX - R_X - XX - R_Y$" design [1]. This "$R_Z -$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

$XX - R_Y - XX - R_Z$" architecture is experimentally found to enable our generator to achieve a more balanced comparability with the proposed HE quantum-driven discriminator. As will be seen, the trained generator in HyperKING significantly outperforms other non-generative quantum or classical AIs, demonstrating the core advantage of HyperKING compared to HyperQUEEN. To further validate this fact, we replace the original generator of HyperKING (i.e., the architecture proposed in Fig. 3) with HyperQUEEN (i.e., the architecture proposed in [1]) while keeping the same training settings (to be detailed later), and we term this approach "HyperQUEEN-QAL." The experiments show that the HyperQUEEN-QAL achieves an improvement by approximately 0.6 dB of peak signal-to-noise ratio (PSNR) over HyperQUEEN; HyperKING further surpasses HyperQUEEN-QAL by around 2 dB in PSNR, substantiating the effectiveness of the "$R_Z - XX - R_Y - XX - R_Z$" architecture and QAL.

Another benchmark HDIP leverages the deep network itself as a specific regularization scheme to take advantage of the deep image network regularization, and extend this idea to the hyperspectral domain [40]. Extending from HDIP, the R-DLRHyIn method not only performs the deep prior but also takes sparse outliers into account, accordingly developing an innovative robust loss function for the low-rank restoration network [41]. HTNN-FFT develops a convex low-rank tensor model for high-order tensor completion based on tensor singular value decomposition (t-SVD) and Tucker nuclear norm (TNN) [39]. As for LRTDTV [38], the global spectral correlation is preserved by low-rank tensor Tucker decomposition, and an anisotropic total-variation regularization is incorporated to further characterize the image smoothness.

In this experiment, we employ a small training dataset that contains only 480 images with spatial size of 128 × 128 pixels collected from NASA's AVIRIS sensor [42]. Besides, we remove those water-vapor absorption spectral bands from original images [1], followed by using the remaining 172 bands as reference images to obtain the training and testing data. To further exhibit the feasibility of our quantum-based generative AI in the SRS area, we also perform the restoration task on the real-world NASA's Earth Observation-1 (EO-1) satellite image.

As for the training settings of the proposed HyperKING, the batch size and the number of training epochs are set to 16 and 2300, respectively. The loss functions of the hybrid discriminator and hybrid generator are presented in Section III-B. Besides, we suggest the following training strategy for the proposed HyperKING to have better adversarial learning procedure. We train the hybrid generator first, and then alternatively train the hybrid discriminator and hybrid generator with the alternating period set to 60. Also, during the HyperKING training stage, the nonstationary discriminator loss may lead to unstable training procedure [43], especially when we utilize a momentum-based optimizer (e.g., Adam [44]). Therefore, we employ the RMSProp optimizer [37] with a learning rate of 0.01, which has conducted a promising capability in those highly non-stationary optimization problems [45].

To understand the applicability of the proposed hybrid quantum-classical framework, we also discuss several predominant quantum computing environments, thereby providing insight of our programming strategy for future researchers. Initially, as quantum devices have yet to be commercialized, the various quantum libraries, such as PennyLane [34], Qiskit [46], Torch Quantum Library [47], and Tensorflow quantum [48], are used to facilitate the quantum-based algorithms. Among these platforms, PennyLane and Qiskit would be regarded as the most representative approaches for quantum programming, where PennyLane excels for those automatically differentiable algorithms; Qiskit, developed by IBM, facilitates seamless integration for the open quantum cloud resources (e.g., IBMQ [49]). However, some literature [50] has experimentally demonstrated that PennyLane significantly outperforms Qiskit in computational efficiency. Furthermore, PennyLane's lightning suite, PennyLane Lightning, provides efficient simulator to integrate the GPUs built by NVIDIA or AMD, leading to the number of qubits on a single quantum node up to 30, or even 40. Numerous training-based quantum algorithms [15], [51], [52] also adopt PennyLane for quantum programming. Thus, we consider PennyLane as the quantum programming platform for implementing and training the proposed HyperKING on large-scale data (such as HSIs) to enhance the efficiency; more details of software environments are summarized in the following paragraph. As for the hardware deployment, since our framework is a pretrained approach, it allows users to train the HyperKING through PennyLane, followed by deploying the trained/optimized quantum gate parameters on the physical quantum device (e.g., IBMQ [49]). The details of the integration of quantum software and hardware can be found in [50]. In summary, the proposed HyperKING is highly suitable to quantum environments, even on the larger-scale data inference.

The software environments are Python 3.10.9, PennyLane 0.25.1, Pytorch 2.1.1, and MATLAB R2023a. We leverage computational equipment composed of an NVIDIA RTX 3090 GPU, AMD Ryzen 5950X CPU with 3.40-GHz speed, and 24-GB RAM to train the proposed HyperKING, and all the methods are tested on this equipment. The comparability analysis in Section III-B is then conducted on the aforementioned computational equipment (for HyperKING), as well as the additional equipments composed of NVIDIA RTX 2080Ti GPU, Intel i9-10900X CPU with 3.70-GHz speed, and 12-GB RAM.

B. Comparability Analysis

In this section, we show that the proposed hybrid discriminator has comparable ability for adversarially upgrading the hybrid generator. We also show that this is not a coincidence. If the core quantum EM module deployed on the hybrid discriminator becomes more sophisticated or more simplified, the adversarial learning may not be smoothly done. As mentioned in Section II, the function of the hybrid generator G(·) is to generate a plausible image to cheat the hybrid discriminator, so we naturally adopt the modified entropy loss function [36]

$$\mathcal{L}_{\mathrm{G}} = \|\mathcal{I}_{\mathrm{real}} - \mathcal{I}_{\mathrm{fake}}^{(\mathrm{G})}\|_{\mathrm{S}} + \lambda \ \log\left(1 - \mathrm{D}\left(\mathcal{I}_{\mathrm{fake}}^{(\mathrm{G})}\right) + \delta\right) \quad (2)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

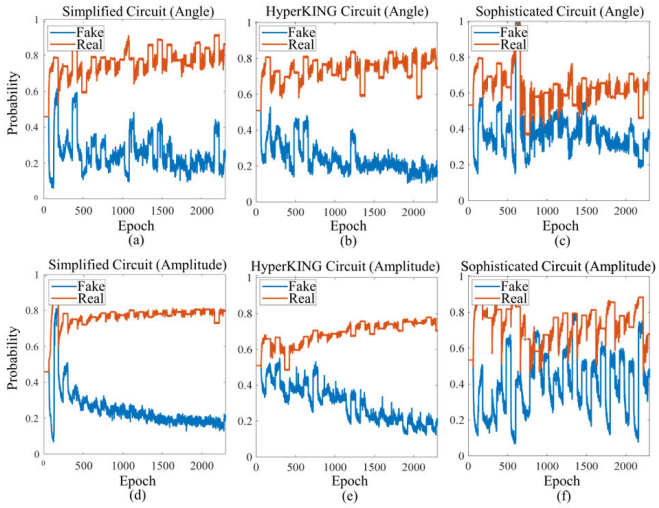


Fig. 7. Probability curves (defined in Fig. 5) for the proposed HyperKING framework and its variations, including the more simplified or more sophisticated quantum EM module (the core in our hybrid discriminator). We also investigate both the amplitude and angle encoding schemes for better understanding of the QAL. (a) Simplified angle-encoding circuit. (b) HypeKING angle-encoding circuit. (c) Sophisticated angle-encoding circuit. (d) Simplified amplitude-encoding circuit. (e) HypeKING amplitude-encoding circuit. (f) Sophisticated amplitude-encoding circuit.

where $\|\cdot\|_S$ is the smoothed ℓ-1 norm [53] to mitigate the outlier effect, $\mathcal{I}_{\text{real}}$ is the real image, $\mathcal{I}_{\text{fake}}^{(G)}$ is the fake image generated by $G(\cdot)$, $\lambda := 0.01$, $\delta := 10^{-8}$, and $D(\cdot)$ is the hybrid discriminator, respectively. Similarly, we adopt

$$\mathcal{L}_D = -\left[\log\left(D(\mathcal{I}_{\text{real}}) + \delta\right) + \log\left(1 - D\left(\mathcal{I}_{\text{fake}}^{(G)}\right) + \delta\right)\right] \quad (3)$$

as the entropy loss for training the quantum discriminator.

Accordingly, a successful QAL is expected to have the following attributes. The hybrid discriminator needs to force the hybrid generator to become more and more stronger. However, when the discriminator is too strong, the generator may give up to become stronger to be able to cheat the discriminator. On the contrary, the hybrid generator may not evolve to a stronger state, if the hybrid discriminator is too weak or too easy to be cheated. Now, we analyze the adversarial curves of HyperKING as displayed in Fig. 5, which shows a successful QAL. First, in the zoom-in local region (see the dashed red rectangle in Fig. 5), one can observe the desired separability of the fake and real image curves during the training period of the hybrid discriminator (i.e., the yellow blocks in Fig. 5), meaning that the discriminator can successfully distinguish whether the input images are real or fake. Second, during the training period of the hybrid generator $G(\cdot)$ (i.e., the gray blocks in Fig. 5), the fake image curve increases as expected, since the generator is being trained to become more capable of generating more plausible images to cheat $D(\cdot)$ [see (3)]. The real image curve remains nearly constant because $D(\cdot)$ remains unchanged during the training period of $G(\cdot)$. Besides the above local attributes, from the global range displayed in the upper part of Fig. 5, one can clearly observe the global separability between the two probability curves, indicating that the QAL successfully helps the discriminator to gradually evolve to stronger states. In summary, a successful QAL is expected to have the above local and global attributes. We remark that due to the inevitable randomness like data shuffle, the QAL procedure may not always exhibit these attributes, but under the aforementioned two sets of computer facilities (see Section III-A), about 70% successful rate for the adversarial learning has been observed. As a side remark for future users of HyperKING (probably with different applications), both the learning rate and the initial seed could also affect the rate of success [54].

To understand that the successful QAL achieved in Fig. 5 is not easy, we show that such a balance may be broken if $G(\cdot)$ or $D(\cdot)$ gets too strong or too weak. To this end, the hybrid discriminator is revised to have more sophisticated or more simplified quantum structures, while the classical parts remain unchanged. Specifically, we enhance the 4-qubit EM module $\text{EM}(bcd|a)$ in the hybrid discriminator (see Fig. 4), to the stronger 8-qubit version $\overline{\text{EM}}(\boldsymbol{v}_{-k}^{(8)}|k)$ $(k := 0, \ldots, 7)$ and to the weakened 2-qubit version $\overline{\text{EM}}(\boldsymbol{v}_{-k}^{(2)}|k)$ $(k := 0, 1)$, where the notation $\boldsymbol{v}_{-k}^{(8)}$ is sequentially composed of all the qubits "7, 6, 5, . . . , 1, 0" but "k" (and $\boldsymbol{v}_{-k}^{(2)}$ is similarly defined). For example, we have $\overline{\text{EM}}(\boldsymbol{v}_{-4}^{(8)}|4) = \overline{\text{EM}}(7653210|4)$, denoting the quantum module that uses qubit "4" to sequentially control the other seven qubits "7 653 210" through the CRX gates, as graphically illustrated in Fig. 6. Accordingly, the more sophisticated quantum structure is designed by replacing the four EM modules of $D(\cdot)$ with the eight modules "$\overline{\text{EM}}(\boldsymbol{v}_{-7}^{(8)}|7)$, $\overline{\text{EM}}(\boldsymbol{v}_{-6}^{(8)}|6)$, . . . , $\overline{\text{EM}}(\boldsymbol{v}_{-0}^{(8)}|0)$," and the more simplified structure is to replace the EM modules with the two modules "$\underline{\text{EM}}(\boldsymbol{v}_{-1}^{(2)}|1)$, $\underline{\text{EM}}(\boldsymbol{v}_{-0}^{(2)}|0)$." Thus, there are in total three modules (i.e., $\overline{\text{EM}}$, EM, and $\underline{\text{EM}}$) and two quantum encoding schemes (i.e., angle encoding and amplitude encoding), leading to a total of six potential combinations. All six combinations have been tested with results collectively summarized in Fig. 7.

One can see that the original HyperKING architecture [i.e., Fig. 7(e)], which adopts the amplitude encoder in $D(\cdot)$ (see Fig. 4), best holds the desired attributes of a successful QAL, among the six cases. Compared to the amplitude encoding scheme [see Fig. 7(d)–(f)], the angle encoding has relatively weak global separability between the real and fake image curves [see Fig. 7(a)–(c)]. The reason is that amplitude encoding encodes exponentially more quantum feature information when compared to the angle encoding [33], meaning that the amplitude encoding scheme corresponds to a much larger receptive field. Also, one can see that when the architecture of the quantum module deployed on the discriminator is too strong/sophisticated [see Fig. 7(c) and (f)], the generator would give up to cheat it, leading to the completely failed QAL. Though this situation has been improved for the case of simplified discriminator architecture [see Fig. 7(a) and (d)], the desired curve attributes are not as clear as those of the original HyperKING [i.e., Fig. 7(e)]. However, one can still see that for the simplified case, the amplitude encoding also exhibits better adversarial learning curves than the angle encoding.

To further emphasize the significance of balanced comparability between the hybrid generator and discriminator, we quantitatively evaluate the performance within the same epoch range under these QALs, as presented in Fig. 7. The details of the landscapes and quantitative metrics used in this experiment can be found in Section III-D. Initially, in the cases of the sophisticated discriminators [i.e., Fig. 7(c) and 7(f)],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

TABLE V
INFERENCE TIME T (SECONDS) OF THE PROPOSED HYPERKING AND OTHER RESTORATION BASELINES

Methods	Time T
LRTDTV [38]	11.920
HDIP [40]	16.412
HTNN-FFT [39]	45.307
R-DLRHyIn [41]	43.475
HyperQUEEN [1]	**0.037**
HyperKING	0.057

they merely achieve approximately 31 dB of PSNR averaged across the landscapes due to the failed QALs. Conversely, the cases of the simplified discriminators [i.e., Fig. 7(a) and 7(d)] improve performance into a PSNR of around 34 dB because their compatibility is relatively balanced compared to the sophisticated cases according to the adversarial curves. This observation is also well aligned with the above analysis for comparability between the hybrid generator and discriminator. Nevertheless, the simplified architectures may not be effective enough to push our hybrid generator to further upgrade. In the cases of the proposed HyperKING [i.e., Fig. 7(b) and 7(e)], they further upgrade the PSNR by approximately 1 dB compared with those simplified cases. On the other hand, the discriminator with amplitude encoding can more effectively force the generator to become stronger than the discriminator with angle encoding; for example, the former achieves an SAM index of 2.5°, whereas the latter only reaches 3.2. This is also consistent with our analysis of the adversarial curves. It is interesting to note that although the adversarial curves presented in Fig. 7(d) seem to be smoother than our original HyperKING [i.e., Fig. 7(e)], the original HyperKING still accomplishes the best quantitative performance. For this observation, we remark that the smooth training curves do not necessarily yield the best performance due to the intractable non-convexity of deep learning [55]. Specifically, an over-smooth training may rapidly converge to an unfavorable local minimum, whereas the slight instability does help the network to escape out of some local optimal regions, hence leading to better results. Also, the simplified discriminator hardly pushes the generator to the optimal equilibrium, as we mentioned.

All in all, the judiciously designed HyperKING has proposed a hybrid discriminator that is comparable to the hybrid generator, and this comparability could be broken when the hybrid discriminator/generator are not carefully balanced. This emphasizes the exquisite structures behind HyperKING. Based on the abovementioned analysis, we recommend that future researchers design the generator and discriminator with an equal number of qubits for balanced comparability. Under the balanced QAL framework, one may consider adopting the amplitude encoding in the discriminator to encode exponentially more quantum feature information, thereby pushing the generator to achieve better results.

C. Qualitative Evaluation

To investigate the qualitative performance of our quantum-based HyperKING when processing SRS images, we test the renowned real-world HSI captured by NASA's EO-1 satellite over Bhilwara of India [56]. The sensor captures the wavelengths ranging from 0.4 to 2.5 μm with a spectral resolution of 10 nm, forming HSIs with 224 spectral bands. The region of interest (ROI) containing 128 × 128 pixels [57] is illustrated in Fig. 8. We remove those water-vapor bands (i.e., 1-7, 61-77, 122-128, 166-178, 217-242 [58]) and use 172 spectral bands in this evaluation. Besides, we select some representative bands with different types of corruption, including band 100, band 99, band 58, and band 57 (see the first column of Fig. 9).

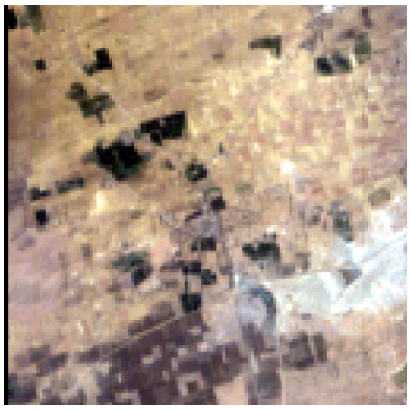


Fig. 8. False color composition [bands (32, 23, 12) as RGB] of the well-known real hyperspectral data over Bhilwara of India.

The restoration results and inference time are, respectively, presented in Fig. 9 and Table V. One can observe that band 99 and band 57 have sparse and dense strip missing patterns, respectively. From Fig. 9, we can see that only HyperQUEEN and the proposed HyperKING have well recovered the corrupted regions without any visual or color distortions. As for band 58, it is the most serious case corresponding to the completely missing pattern. In this scenario, HDIP and R-DLRHyIn may have some unexpected spatial patterns because of the absence of a discriminator, while HTNN-FFT fails to recover this band image. LRTDTV and HyperQUEEN have reasonable recovery results, but LRTDTV does not well align with the ground-truth spatial texture as presented in the ROI (see Fig. 8). The proposed HyperKING exhibits its superiority and well recovers this completely missing band under a fully blind setting.

In summary, the proposed HyperKING achieves the best qualitative performance in the real-world satellite data restoration scenario. Notably, the inference time of quantum-based baseline (i.e., HyperQUEEN) and our quantum-based generative AI (i.e., HyperKING) is several orders of magnitude faster than other classical baseline methods (see Table V), demonstrating the feasibility of the real-time satellite signal processing via quantum machine intelligence (quantum-based generative AI). Note that as the experimental settings are exactly the same as in [1], the experimental results of HyperQUEEN are replicated from [1]. Beyond qualitative evaluation, we will also perform a quantitative evaluation in Section III-D using all the four representative landscapes (i.e., city, farm, lake, mountain) and Wald's protocol [36], [59] for a comprehensive comparison.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

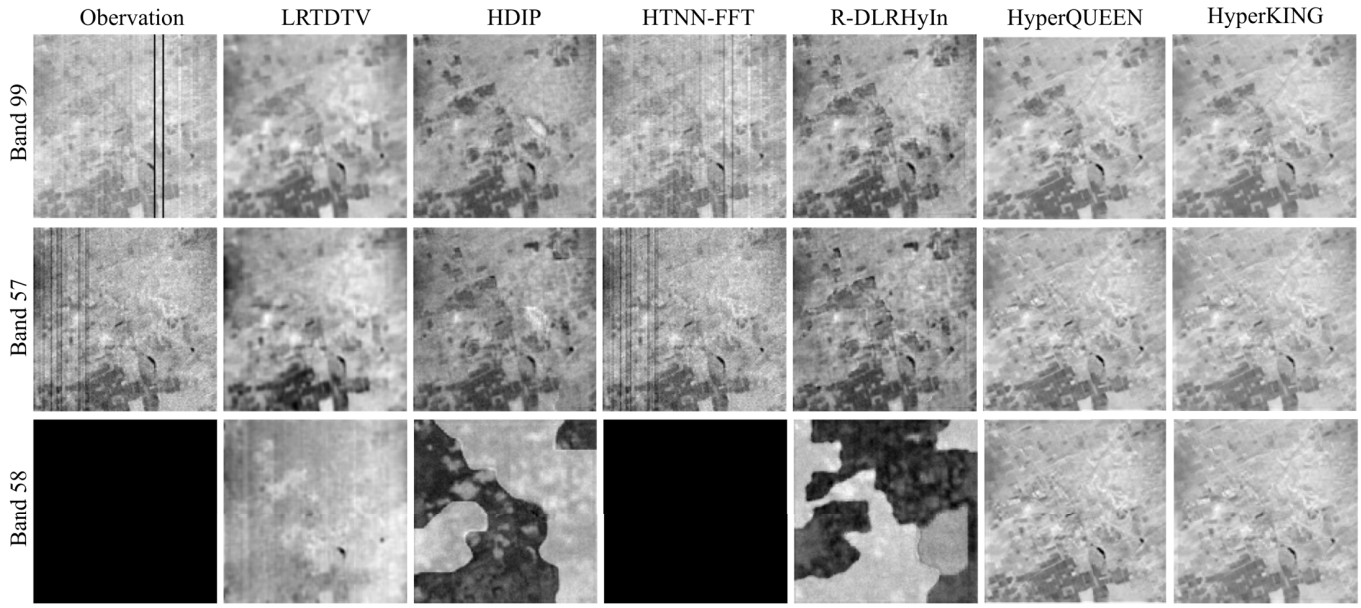


Fig. 9. Visual comparisons of restored NASA's EO-1 Hyperion satellite images captured over Bhilwara of India. The representative spectral bands regarding sparse corruption (i.e., band 99), dense corruption (i.e., band 57), and serious corruption (i.e., band 58) are leveraged for this qualitative evaluation.

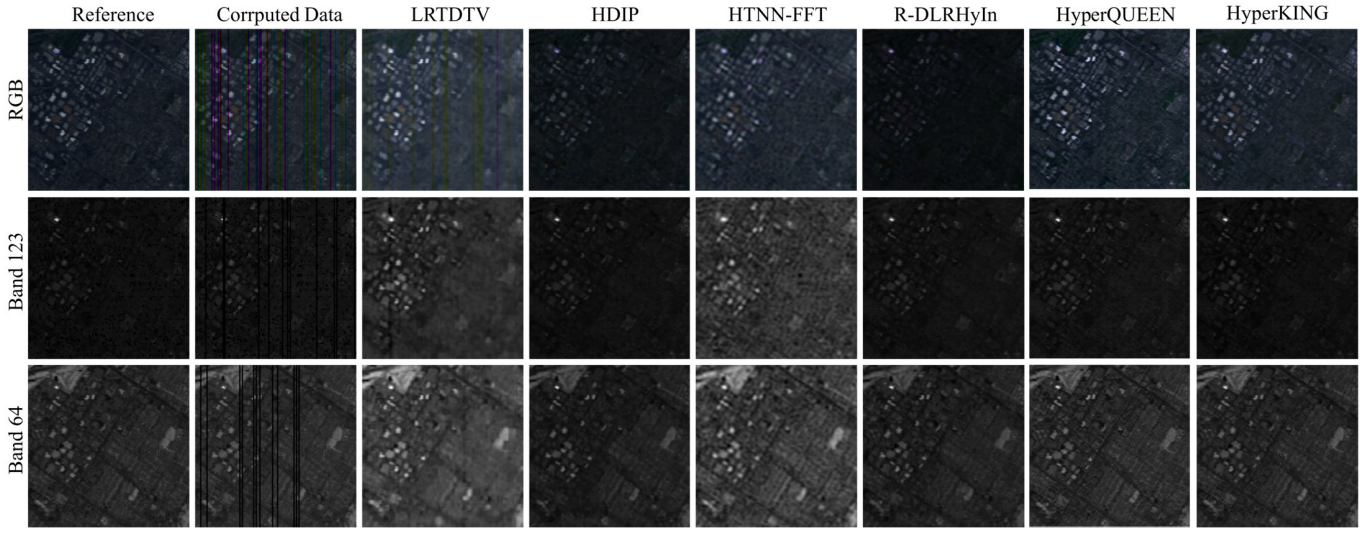


Fig. 10. Restoration of NASA's AVIRIS data captured over city landscape. Representative spectral bands, including the true-color composition bands, band 123, and band 64, are displayed.

TABLE VI

QUANTITATIVE EVALUATION OF HYPERKING AND OTHER BASELINES OVER CITY LANDSCAPE

Methods	PSNR (↑)	SAM (↓)	RMSE (↓)	SSIM (↑)	Time T
LRTDTV [38]	28.721	5.265	0.015	0.915	21.285
HDIP [40]	36.702	6.224	0.014	0.941	21.170
HTNN-FFT [39]	30.468	5.369	0.015	0.909	36.379
R-DLRHyIn [41]	37.042	6.038	0.014	0.943	48.134
HyperQUEEN [1]	38.918	3.557	0.008	0.979	**0.026**
HyperKING	**41.971**	**3.049**	**0.007**	**0.984**	0.057

TABLE VII

QUANTITATIVE EVALUATION OF HYPERKING AND OTHER BASELINES OVER FARM LANDSCAPE

Methods	PSNR (↑)	SAM (↓)	RMSE (↓)	SSIM (↑)	Time T
LRTDTV [38]	29.787	**1.977**	0.017	0.929	21.039
HDIP [40]	28.032	5.161	0.022	0.925	16.031
HTNN-FFT [39]	31.568	2.071	0.015	0.940	36.550
R-DLRHyIn [41]	29.204	4.500	0.020	0.935	4.514
HyperQUEEN [1]	31.014	3.538	0.017	0.967	**0.025**
HyperKING	**33.461**	2.802	**0.013**	**0.981**	0.057

D. Quantitative Evaluation

Next, we further conduct quantitative evaluations based on well-known quantitative metrics, i.e., PSNR [60], spectral angle mapper (SAM) [36], root-mean-square error (RMSE) [1], and structural similarity (SSIM) [61]. Besides, we select the four representative landscapes from NASA's AVIRIS data (see Section III-A), i.e., city, farm, lake, and mountain, whose

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

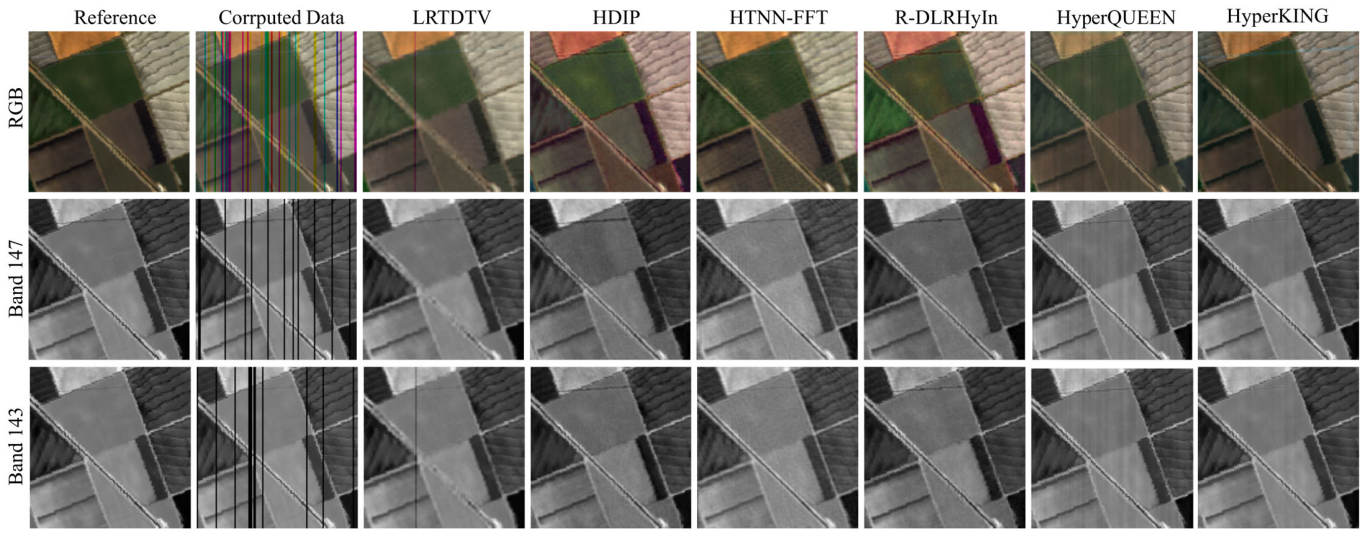


Fig. 11. Restoration of NASA's AVIRIS data captured over farm landscape. Representative spectral bands, including the true-color composition bands, band 147, and band 143, are displayed.

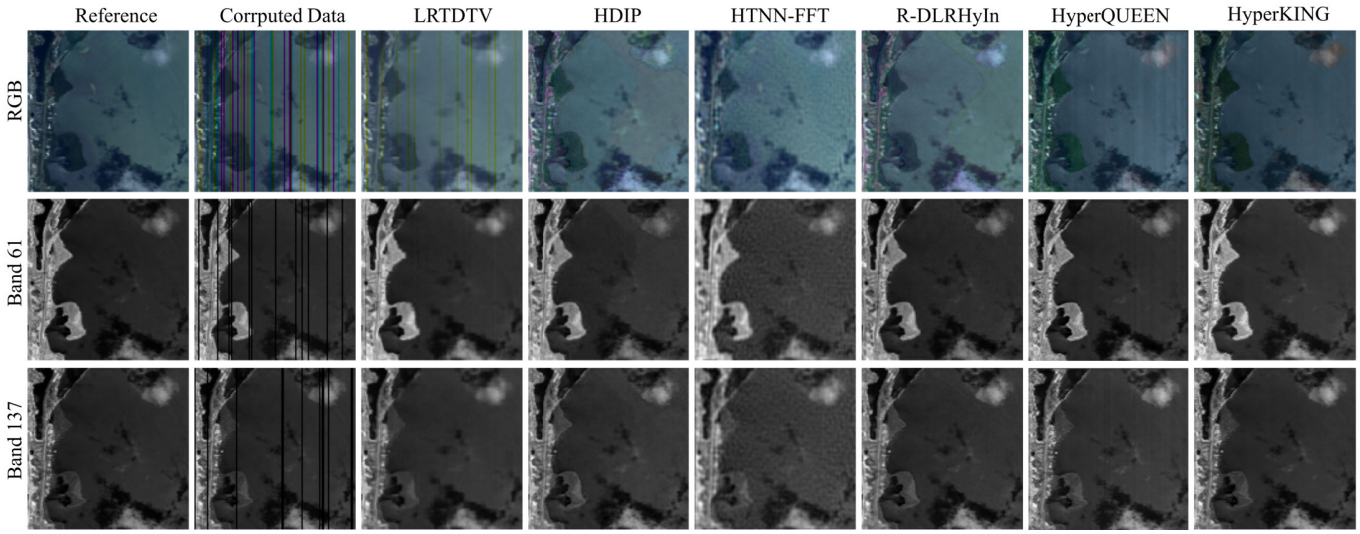


Fig. 12. Restoration of NASA's AVIRIS data captured over lake landscape. Representative spectral bands, including the true-color composition bands, band 61, and band 137, are displayed.

TABLE VIII

QUANTITATIVE EVALUATION OF HYPERKING AND OTHER BASELINES OVER LAKE LANDSCAPE

Methods	PSNR (↑)	SAM (↓)	RMSE (↓)	SSIM (↑)	Time T
LRTDTV [38]	30.971	3.791	0.016	0.973	21.611
HDIP [40]	30.485	5.702	0.017	0.966	16.287
HTNN-FFT [39]	28.470	**2.880**	0.014	0.956	38.312
R-DLRHyIn [41]	31.939	4.226	0.013	0.974	43.904
HyperQUEEN [1]	30.794	3.468	0.014	0.985	**0.025**
HyperKING	**33.604**	2.899	**0.010**	**0.991**	0.057

TABLE IX

QUANTITATIVE EVALUATION OF HYPERKING AND OTHER BASELINES OVER MOUNTAIN LANDSCAPE

Methods	PSNR (↑)	SAM (↓)	RMSE (↓)	SSIM (↑)	Time T
LRTDTV [38]	31.086	1.826	0.018	0.929	20.695
HDIP [40]	27.133	4.692	0.031	0.854	16.227
HTNN-FFT [39]	32.012	1.855	0.016	0.937	36.763
R-DLRHyIn [41]	27.929	4.257	0.028	0.868	43.359
HyperQUEEN [1]	32.984	1.731	0.016	0.965	**0.026**
HyperKING	**35.990**	**1.376**	**0.010**	**0.984**	0.06

ROIs are illustrated in the first column of Figs. 10–13, respectively. In this evaluation, we employ the dense stripes missing that are widely seen in SRS images to conduct the tensor completion task. In detail, we randomly remove about 10% stripes in each spectral band of the corresponding reference (see the second columns in Figs. 10–13). Notably, although the corrupted stripes are padded with zeros, their corrupted patterns differ across each band, resulting in color stripes instead of black stripes in our tensor completion task. Note that in real-world sensor array systems, color stripes are more frequently seen as the position where the stripe occurs (i.e., the position where the sensor is broken) differs across each

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

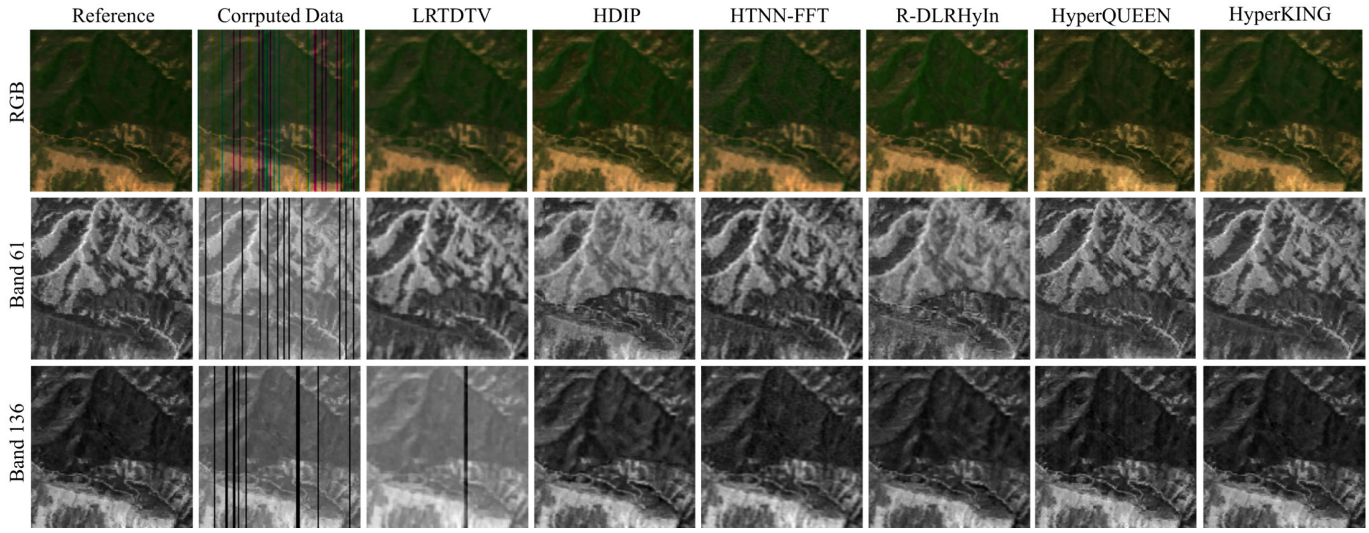


Fig. 13. Restoration of NASA's AVIRIS data captured over mountain landscape. Representative spectral bands, including the true-color composition bands, band 61, and band 136, are displayed.

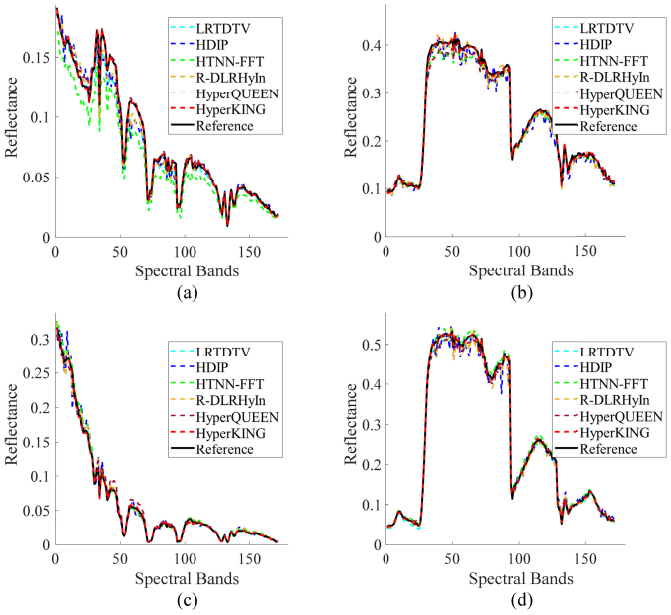


Fig. 14. Qualitative comparisons for the spectral profiles over the (a) city, (b) farm, (c) lake, and (d) mountain landscapes.

band. Then, the quantitative results of city, farm, lake, and mountain are organized, respectively, in Tables VI–IX.

First of all, one can observe that the proposed HyperKING outperforms other baselines over city and mountain ROIs in terms of all the PSNR/SAM/RMSE/SSIM. Remarkably, the PSNR values over the city and mountain landscapes of HyperKING (see Tables VI and IX) are higher than the strongest quantum-based benchmark (i.e., HyperQUEEN) by around 3 dB. This is a direct evidence of the superiority of the QAL. Furthermore, the SAM and SSIM of the proposed HyperKING are also improved, again demonstrating the feasibility of QAL. Moreover, the proposed HyperKING also achieves the strongest quantitative performances in terms of PSNR/RMSE/SSIM over farm and lake landscapes (see Tables VII and VIII). LRTDTV achieves the best SAM in farm landscapes, as it leverages the tensor Tucker decomposition to preserve the inherent global spectral correlation. On the other hand, HTNN-FFT conducts a high-order convex tensor low-rank model, and this low-rank modeling may lead to such a superior spectral shape preservation, as indicated by the lowest SAM value over lake landscape. All in all, the proposed HyperKING still exhibits superior performance over all the four representative remote sensing landscapes. Furthermore, compared with the pioneering quantum-based network, i.e., HyperQUEEN, the performance of HyperKING is significantly enhanced thanks to the QAL. Also, the inference time for the quantum-based methods shows that real-time satellite data restoration is feasible through quantum-based machine intelligence (e.g., HyperQUEEN) or quantum generative AI (e.g., our HyperKING).

To have better understanding, the corresponding visual results based on these four landscapes are also analyzed. In Fig. 10, one can observe that the results of the HDIP-based approach (e.g., R-DLRHyIn) have very slight chroma distortion. On the other hand, LRTDTV restores the corruption with a slight blurring effect, which is similar to the real-world data test on the Bhilwara region (see Section III-C). Only quantum-based methods (i.e., HyperQUEEN and HyperKING) yield promising visual quality in the city ROI, which echoes the above quantitative analysis (see Table VI). However, in the farm landscape (see Fig. 11), the results of HyperQUEEN have slight stripe texture, and this has been significantly improved by HyperKING, as the proposed QAL can surge the QUEEN-based hybrid generator to be upgraded to a new level, echoing the result presented in Section III-B. The hybrid generator and hybrid discriminator compete against each other hence achieving even superior performance comparing to the case of independently training a single network. On the other hand, we remark that HTNN-FFT also achieves promising qualitative performance. In particular, the upper-right region in the RGB composition of HTNN-FFT is relatively close to the reference, demonstrating its effectiveness in spectral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

preserving. However, the unfavorable grid-like patterns in the middle region of HTNN-FFT reveal its limitation in restoring the spatial details. These observations are well aligned with the quantitative evaluations (see Table VII); for example, HTNN-FFT achieves promising SAM values, exhibiting its effectiveness in preserving spectral relations. In lake landscape (see Fig. 12), the slight stripe observed in the result of HyperQUEEN is also eliminated by HyperKING. As for the mountain landscapes (see Fig. 13), we can observe that the result of R-DLRHyIn has chroma distortion in the RGB composition while the overall performance is still promising.

On the other hand, preserving the spectral profiles is also critical to facilitate SRS applications. To this end, the qualitative comparison of the spectral profiles is visualized in Fig. 14. We first analyze the spectral profile over the city landscape in Fig. 14(a), which substantiates that the restored spectral profile of HTNN-FFT has serious deviations from the reference. Moreover, the spectral profile of other benchmarks (e.g., R-DLRHyIn and HDIP) appears to be noisy, whereas the true spectral profile is expected to be smooth. This adverse phenomenon is more significant over the farm landscape, as shown in Fig. 14(b). Notably, only the quantum-based methods (i.e., HyperQUEEN and HyperKING) successfully achieve promising spectral profiles, outperforming the other classical approaches. As for the spectral profiles of the lake and mountain landscapes, all approaches seem to have relatively favorable spectral profiles, while HyperKING further suppresses the noisy effect observed in the classical baselines hence accomplishing the most accurate restorations. All in all, thanks to the effective QAL, the proposed HyperKING has the best qualitative and quantitative performances.

E. Hyperspectral Unmixing Evaluation

As the key value of HSI lies within its material identifiability, the restored images should be able to yield correct hyperspectral signatures (a.k.a. endmembers) for the subsequent classification task. Thus, a critical BSS technique [62], [63] in the SRS field, particularly named HU [64], extracts the endmembers from the target HSI. Specifically, a pixel in a remotely sensed HSI is usually a mixture of several pure endmembers (i.e., material sources). So, we will need BSS technique to unmix the hyperspectral mixture for recovering the pure endmembers, and this is the so-called HU technique in the SRS area. We also remark that the HU technique, serving as the fundamental SRS task, empowers numerous SRS applications, such as agricultural monitoring [65], urban planning [66], and anomaly detection [20]. Accordingly, the applicability of the restoration approaches in different SRS applications can be efficiently evaluated by examining whether the restorations from HyperKING can facilitate precise HU, rather than applying them on individual SRS tasks separately. More specifically, if a restored image cannot yield satisfactory HU results, the restoration method is considered less practical, and vice versa. Therefore, we will use HU result to evaluate the restoration capability of HyperKING, and we will adopt the widely used metric ϕ_{en} (i.e., root-mean-square (RMS) spectral angle error) to evaluate the quality of the unmixed endmembers. A lower value of ϕ_{en} corresponds to a better quality of endmembers, and the definition of ϕ_{en} can be found in [67, eq. (32)].

TABLE X
HU EVALUATION IN TERMS OF ϕ_{en} (IN DEGREES) OVER THE FOUR REPRESENTED LANDSCAPES

Methods	City	Farm	Lake	Mountain	Average
LRTDTV [38]	20.375	10.345	9.058	10.207	12.496
HDIP [40]	14.311	13.537	15.086	9.123	13.014
HTNN-FFT [39]	10.409	7.807	4.778	6.529	7.381
R-DLRHyIn [41]	17.836	12.701	6.927	8.156	11.405
HyperQUEEN [1]	9.354	12.274	5.997	7.998	8.905
HyperKING	**8.691**	**5.623**	**4.056**	**5.338**	**5.927**

To demonstrate the effectiveness of the proposed QAL, we employ the fast and theoretically reliable HU algorithm, called hyperplane-based Craig simplex identification (HyperCSI) [67]. Specifically, HyperCSI is applied to separate the hyperspectral pixel mixtures for both the reference image and restored image (restored by a particular method like HyperKING), thereby obtaining the original endmembers and the restored endmembers. Since the endmembers recovered by HyperCSI have been theoretically proven to be the true endmembers under mild conditions, as detailed in [67, Th. 2]. Thus, we fairly consider the original endmembers as the reference endmembers, followed by computing the RMS spectral angle error ϕ_{en} between the reference and restored endmembers, in order to evaluate the restoration performance. In the HU algorithm, HyperCSI, the model order N, radius compression ratio r, and simplex compression ratio η are empirically set as $N := 6$, $r := 1E-8$, and $\eta := 0.9$, respectively. The HU results are summarized in Table X. As one can see, the proposed HyperKING quantum-based generative AI has achieved the best performance (i.e., the smallest ϕ_{en}) for all four investigated landscapes, well echoing the quantitative and qualitative evaluations in Sections III-C and III-D. With the highly comparable quantum adversarial framework, the hybrid generator well restores the corruptions and maintains rather accurate spectral attributes. All in all, this BSS experiment again confirms the effectiveness and feasibility of our HyperKING framework.

F. Hyperspectral Mixed Noise Removal Evaluation

In this section, a denoising evaluation is further conducted to substantiate the effectiveness and reliability of the proposed HyperKING. Given that SRS images are not merely corrupted by the damaged sensor array but also degraded by the unknown complex noise during transportation [68]. Therefore, removing the noise degradation before those critical SRS applications (e.g., classification) becomes an essential step. To this end, we adopt a severe mixed noise corruption, including the dead stripes, Gaussian noise, and impulse noise, for this experiment, and the details of mixed noise components are described hereinafter. First, zero-mean Gaussian noise with a standard deviation being the 10% of the maximum reflectance of the HSIs is added for each band. Subsequently, a total of 1%

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

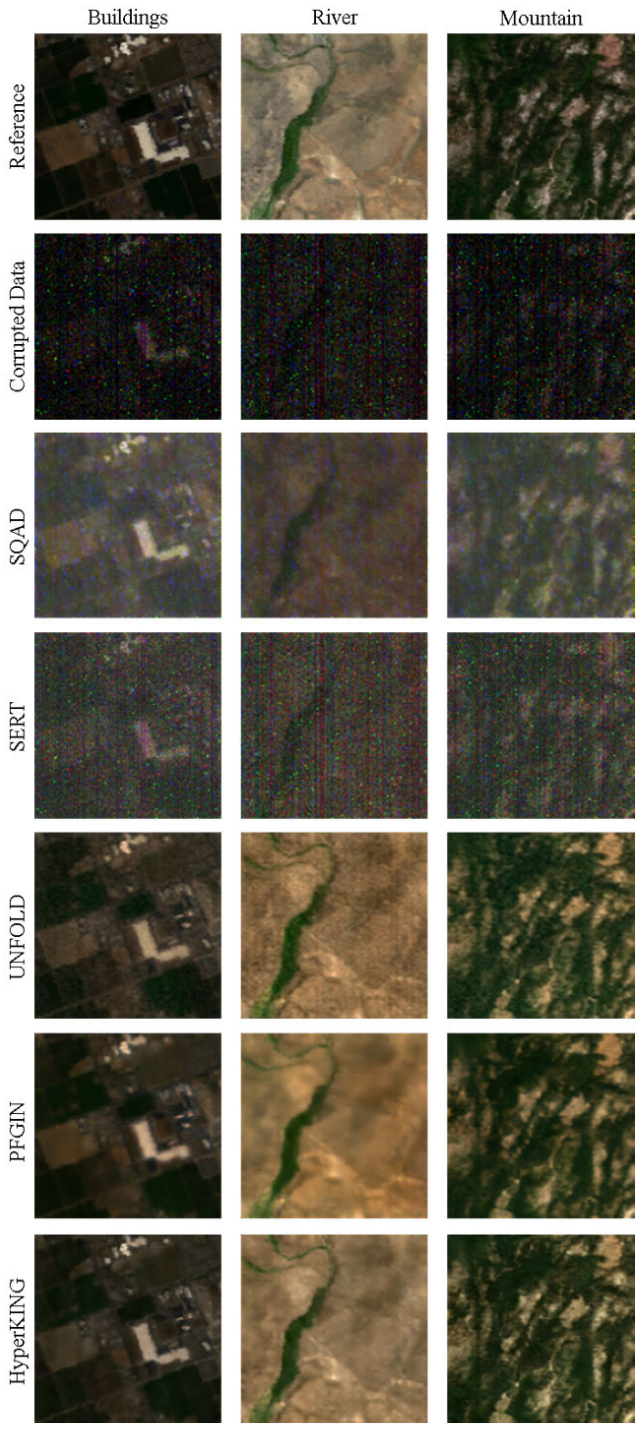


Fig. 15. Qualitative comparisons in true-color composition under the severe mixed noise corruption (i.e., dead stripe, Gaussian noise, and impulse noise) over the representative ROIs in the testing dataset.

elements of the HSI are corrupted by impulse noise. Finally, we randomly add the 10% dead stripe corruption for each band to obtain the mixed noise corrupted HSI. In summary, each band in the HSI in this experiment is corrupted by three distinct types of noise, leading to a highly challenging degradation, as shown in the second row of Fig. 15.

To evaluate the proposed HyperKING, we adopt various types of supervised baselines for this experiment, as the deep learning-based approaches have exhibited their superiority in terms of SRS restoration. As a result, the selected benchmarks are all supervised. Initially, we employ the spatial–spectral quasi-attention recurrent network (abbreviated as SQAD) [68] as the attention-based baseline. In SQAD, the feature extraction is composed of novel quasi-attention recurrent units for local contextual spatial information and spectral correlation. With the success of the self-attention mechanism of transformer architectures, the spectral enhanced rectangle transformer (SERT) for HSI denoising [69] is also considered as the transformer-based method. The spatial rectangle self-attention embedded within SERT first learns the global dependency of spatial patches, and then the spectral enhancement module extracts the low-rank vector to suppress the noise components. Furthermore, an innovative baseline, UNFOLD [70], integrates the strengths of 3-D transformer and 3-D CNN with a U-shaped architecture. Compared to those traditional 2-D networks that overlook the 3-D textures of HSIs, the 3-D transformer and CNN can learn the spectral correlations from the architectural perspective, leading to effective performance. Finally, we also utilize the very recent classical GAN-based approach, i.e., predictive filtering integrated generative HSI inpainting network (PFGIN) [71], as the generative-type baseline for a comprehensive comparison. All methods are trained with the same training dataset mentioned in Section III-A. Besides, we utilize the validation and testing datasets that contain 72 images and 160 images, respectively, for a fair evaluation. All images in these datasets have a spatial size of 128 $\times$ 128 pixels and are also collected from NASA's AVIRIS sensor [42].

We conduct the qualitative comparisons as follows. The representative ROIs selected from the testing dataset, shown in true-color composition, are presented in Fig. 15. Based on the qualitative comparison, the attention-based approaches, such as SQAD and SERT, fail to effectively restore the HSIs. In particular, these approaches can only remove noise effects partially due to the challenging noise corruption, leading to unfavorable restoration outcomes. Next, it can be observed that UNFOLD and PFGIN yield relatively satisfactory restorations, as shown in Fig. 15. However, the noisy patterns are still visible in the results of UNFOLD, which leads to spectral deviations (see river ROI). As for the classical GAN-based method, PFGIN, the results seem to be plausible (but oversmoothed) and lack spatial details. Such an inevitable phenomenon reveals the unreliability of classical generative networks. Compared to these renowned baselines, the proposed quantum-based HyperKING precisely recovers the HSIs from the challenging mixed noise corruption, further demonstrating the effectiveness of QAL.

To further substantiate this fact, we perform the quantitative evaluations as summarized in Table XI. According to the quantitative evaluation, SERT barely restores the noisy HSI, aligning with the qualitative observations. On the other hand, although SQAD accomplishes promising performance in terms of PSNR, its SAM metric is much weaker than UNFOLD and PFGIN. This spectral deviation is consistent with our qualitative analysis as well. Conversely, the proposed HyperKING achieves the best quantitative performance among

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
QUANTITATIVE EVALUATION OF HYPERKING AND OTHER BASELINES UNDER THE SERIOUS MIXED NOISE CORRUPTION. THE QUANTITATIVE METRICS EXHIBIT THE AVERAGED PERFORMANCE OVER THE TESTING DATASET CONTAINING 160 HSIS

Methods	PSNR (↑)	SAM (↓)	RMSE (↓)	SSIM (↑)	Time T
SQAD [68]	27.387	6.536	0.025	0.761	0.095
SERT [69]	18.788	27.698	0.122	0.148	0.028
UNFOLD [70]	26.854	4.431	0.019	0.828	0.031
PFGIN [71]	27.356	3.603	0.020	0.868	**0.013**
HyperKING	**30.217**	**2.940**	**0.016**	**0.915**	0.061

all benchmarks. In particular, its PSNR metric outperforms the strongest baseline (i.e., PFGIN) by approximately 3 dB, substantiating the superiority of QAL. All in all, the proposed HyperKING again outperforms the renowned benchmarks under this highly challenging SRS task.

IV. CONCLUSION AND FUTURE WORKS

We have proposed the "Hyperspectral Knot-like IntelligeNt dIscrimiNator and Generator" (HyperKING) quantum framework, which is the first quantum generative AI that achieves AQIP tasks in hyperspectral SRS. The hybrid generator in HyperKING is designed as the low-rank quantum deep network (QUEEN) with mathematically provable quantum FE for lightweight quantum AI (see Theorem 1), which processes the highly compressed quantum features to save the qubit resources in the near-term quantum computer. The proposed DC mechanism is more effective than those naive approaches (e.g., patchwise computing or PCA compression) used in existing benchmark quantum-based GANs. The QUEEN-based generator also employs the inverse-QC module to map the collapsed quantum image state back to the hyperspectral space. On the other hand, the hybrid discriminator in HyperKING is designed as the CRX-driven HE quantum classifier and has been proven to have comparable ability as the proposed QUEEN-based generator, thereby ensuring that the hybrid discriminator and generator can help each other to get stronger through the QAL. Achieving the comparable ability has not been easy (see Section III-B).

The proposed HyperKING leads to a breakthrough in the area of quantum image processing, as it significantly upgrades the spatial capacity (from 28×28 to 128×128 resolution) and spectral capacity (from grayscale to 172 channel). Our HyperKING also achieves superior satellite data restoration results, such as hyperspectral tensor completion and mixed noise removal, with very fast computational time. Our experiments also show that the hybrid generator/discriminator in HyperKING compete against each other, hence achieving significantly superior performance when compared to the case of independently training a single quantum-based network (e.g., HyperQUEEN). This is an important reason why we choose GAN as our first step toward quantum generative AI; one would pursue quantum-based diffusion models as the next step. Remarkably, HyperKING achieves the above objectives under a fully *blind* setting without needing to know the locations of the damaged hyperspectral pixels. The carefully reported design philosophy behind HyperKING (see Section II) also serves as a practical guide for future investigators to study the quantum generative AI.

Another of our ongoing research lines is to explore more advanced applications of HyperKING, such as hyperspectral object counting and hyperspectral classification, for both of which HU-driven identification [18] is critical. To this end, we further consider the small-data (or even single-data) QAL as the future work, not only echoing the relatively scarce training samples for SRS classification (compared to RGB classification) but echoing the limited qubit resources in the near-term quantum computers. Although our experiment shows that HyperKING trained using very few data (i.e., only 480 HSIs, each with 128×128 pixels) still exhibits high performance for various SRS tasks, its general applicability would be further explored in the future. We expect that the proposed HyperKING framework, having already proved FE (implying the capability of expressing numerous functionalities for diverse applications), will further achieve other important SRS tasks through just small-data (or single-data) learning, probably requiring customized QAL or upgraded hybrid generator (and its counterpart discriminator with comparable ability, see Fig. 7).

ACKNOWLEDGMENT

The authors would like to thank the National Center for Theoretical Sciences (NCTS) and the National Center for High-performance Computing (NCHC) for providing the computing resources.

REFERENCES

[1] C.-H. Lin and Y.-Y. Chen, "HyperQUEEN: Hyperspectral quantum deep network for image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511720.
[2] P. B. Upama et al., "Evolution of quantum computing: A systematic survey on the use of quantum computing tools," 2022, *arXiv:2204.01856*.
[3] R. D. Camino, R. State, L. Montero, and P. Valtchev, "Finding suspicious activities in financial transactions and distributed ledgers," in *Proc. IEEE Int. Conf. Data Mining Workshops (ICDMW)*, Nov. 2017, pp. 787–796.
[4] M. Boyer, G. Brassard, P. Høyer, and A. Tapp, "Tight bounds on quantum searching," *Fortschritte der Physik*, vol. 46, nos. 4–5, pp. 493–505, Jun. 1998.
[5] P. W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer," *SIAM Rev.*, vol. 41, no. 2, pp. 303–332, Jan. 1999.
[6] J. Tian et al., "Recent advances for quantum neural networks in generative learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12321–12340, Oct. 2023.
[7] C.-H. Lin, T.-H. Lin, and J. Chanussot, "Quantum information-empowered graph neural network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537615.
[8] C.-H. Lin, P.-W. Tang, and A. R. Huete, "Quantum feature-empowered deep classification for fast mangrove mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5400313.
[9] C.-H. Lin and J.-T. Lin, "PRIME: Blind multispectral unmixing using virtual quantum prism and convex geometry," 2024, *arXiv:2407.15358*.
[10] G. Gao, Q. Liu, and Y. Wang, "Counting from sky: A large-scale data set for remote sensing object counting and a benchmark method," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 3642–3655, May 2021.
[11] H. Wang et al., "Hierarchical kernel interaction network for remote sensing object counting," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5604113.
[12] J. Gui, Z. Sun, Y. Wen, D. Tao, and J. Ye, "A review on generative adversarial networks: Algorithms, theory, and applications," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 3313–3332, Apr. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

[13] H.-L. Huang et al., "Experimental quantum generative adversarial networks for image generation," *Phys. Rev. Appl.*, vol. 16, no. 2, Aug. 2021, Art. no. 024051.

[14] N.-R. Zhou, T.-F. Zhang, X.-W. Xie, and J.-Y. Wu, "Hybrid quantum–classical generative adversarial networks for image generation via learning discrete distribution," *Signal Process., Image Commun.*, vol. 110, Jan. 2023, Art. no. 116891.

[15] S. L. Tsang, M. T. West, S. M. Erfani, and M. Usman, "Hybrid quantum–classical generative adversarial network for high-resolution image generation," *IEEE Trans. Quantum Eng.*, vol. 4, pp. 1–19, 2023.

[16] S. A. Stein et al., "QuGAN: A quantum state fidelity based generative adversarial network," in *Proc. IEEE Int. Conf. Quantum Comput. Eng. (QCE)*. Broomfield, CO, USA: IEEE, Oct. 2021, pp. 71–81.

[17] I. Marvian, "Restrictions on realizable unitary operations imposed by symmetry and locality," *Nature Phys.*, vol. 18, no. 3, pp. 283–289, Mar. 2022.

[18] C.-H. Lin, W.-K. Ma, W.-C. Li, C.-Y. Chi, and A. Ambikapathi, "Identifiability of the simplex volume minimization criterion for blind hyperspectral unmixing: The no-pure-pixel case," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5530–5546, Oct. 2015.

[19] P. Ball, "First quantum computer to pack 100 qubits enters crowded race," *Nature*, vol. 599, no. 7886, p. 542, Nov. 2021.

[20] S.-S. Young, C.-H. Lin, and Z.-C. Leng, "Unsupervised abundance matrix reconstruction transformer-guided fractional attention mechanism for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–15, Aug. 2024.

[21] T.-H. Lin and C.-H. Lin, "Hyperspectral change detection using semi-supervised graph neural network and convex deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515818.

[22] D. Fudenberg and J. Tirole, *Game Theory*. Cambridge, MA, USA: MIT Press, 1991.

[23] S. Sim, P. D. Johnson, and A. Aspuru-Guzik, "Expressibility and entangling capability of parameterized quantum circuits for hybrid quantum-classical algorithms," *Adv. Quantum Technol.*, vol. 2, no. 12, Dec. 2019, Art. no. 1900070.

[24] I. J. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[25] G. E. Crooks. (2020). *Gates, States, and Circuits*. [Online]. Available: https://threeplusone.com/pubs/on_gates.pdf

[26] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*. Cambridge, U.K.: Cambridge Univ. Press, 2010.

[27] C.-Y. Chi, W.-C. Li, and C.-H. Lin, *Convex Optimization for Signal Processing and Communications: From Fundamentals to Applications*. Boca Raton, FL, USA: CRC Press, 2017.

[28] G. Cooper, "Examining science education in ChatGPT: An exploratory study of generative artificial intelligence," *J. Sci. Educ. Technol.*, vol. 32, no. 3, pp. 444–452, 2023.

[29] V. Lempitsky, A. Vedaldi, and D. Ulyanov, "Deep image prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9446–9454.

[30] C.-H. Lin and J. M. Bioucas-Dias, "An explicit and scene-adapted definition of convex self-similarity prior with application to unsupervised Sentinel-2 super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3352–3365, May 2020.

[31] C.-H. Lin and J. M. Bioucas-Dias, "Nonnegative blind source separation for ill-conditioned mixtures via John ellipsoid," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 5, pp. 2209–2223, May 2021.

[32] M. Weigold, J. Barzen, F. Leymann, and M. Salm, "Encoding patterns for quantum algorithms," *IET Quantum Commun.*, vol. 2, no. 4, pp. 141–152, Dec. 2021.

[33] F. Yan and S. E. Venegas-Andraca, *Quantum Image Processing*. Cham, Switzerland: Springer, 2020.

[34] V. Bergholm et al., "PennyLane: Automatic differentiation of hybrid quantum-classical computations," 2018, *arXiv:1811.04968*.

[35] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2019, pp. 1–12.

[36] C.-H. Lin, Y.-C. Lin, and P.-W. Tang, "ADMM-ADAM: A new inverse imaging framework blending the advantages of convex optimization and deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

[37] T. Tieleman and G. Hinton, "Lecture 6.5-RMSPROP: Divide the gradient by a running average of its recent magnitude," *COURSERA, Neural Netw. Mach. Learn.*, vol. 4, no. 2, pp. 26–31, 2012.

[38] Y. Wang, J. Peng, Q. Zhao, Y. Leung, X.-L. Zhao, and D. Meng, "Hyperspectral image restoration via total variation regularized low-rank tensor decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1227–1243, Apr. 2018.

[39] W. Qin, H. Wang, F. Zhang, J. Wang, X. Luo, and T. Huang, "Low-rank high-order tensor completion with applications in visual data," *IEEE Trans. Image Process.*, vol. 31, pp. 2433–2448, 2022.

[40] O. Sidorov and J. Y. Hardeberg, "Deep hyperspectral prior: Single-image denoising, inpainting, super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 3844–3851.

[41] K. Faghih Niresi and C.-Y. Chi, "Robust hyperspectral inpainting via low-rank regularized untrained convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[42] *AVIRIS Free Standard Data Products*. Accessed: Nov. 2, 2019. [Online]. Available: http://aviris.jpl.nasa.gov/html/aviris.freedata.html

[43] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein GAN," 2017, *arXiv:1701.07875*.

[44] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent.*, San Diego, CA, USA, Dec. 2014, pp. 1–13.

[45] V. Mnih et al., "Asynchronous methods for deep reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1928–1937.

[46] A. Javadi-Abhari et al., "Quantum computing with qiskit," 2024, *arXiv:2405.08810*.

[47] H. Wang et al., "QuantumNAS: Noise-adaptive search for robust quantum circuits," in *Proc. IEEE Int. Symp. High-Perform. Comput. Archit. (HPCA)*, Seoul, (South) Korea, Apr. 2022, pp. 692–708.

[48] M. Broughton et al., "TensorFlow quantum: A software framework for quantum machine learning," 2020, *arXiv:2003.02989*.

[49] U. Alvarez-Rodriguez, M. Sanz, L. Lamata, and E. Solano, "Quantum artificial life in an IBM quantum computer," *Sci. Rep.*, vol. 8, no. 1, p. 14793, Oct. 2018.

[50] M. Haghparast, E. Moguel, J. Garcia-Alonso, T. Mikkonen, and J. M. Murillo, "Innovative approaches to teaching quantum computer programming and quantum software engineering," in *Proc. IEEE Int. Conf. Quantum Comput. Eng. (QCE)*, Montreal, QC, Canada, Sep. 2024, pp. 251–255.

[51] A. Pandey and B. Rudra, "Deepfake audio detection using quantum learning models," in *Proc. IEEE Middle East Conf. Commun. Netw. (MECOM)*, Abu Dhabi, United Arab Emirates, Nov. 2024, pp. 1–6.

[52] Y. Kwak, W. J. Yun, S. Jung, J.-K. Kim, and J. Kim, "Introduction to quantum reinforcement learning: Theory and PennyLane-based implementation," in *Proc. Int. Conf. Inf. Commun. Technol. Converg. (ICTC)*, Oct. 2021, pp. 416–420.

[53] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[54] D. Picard, "Torch.manual_seed(3407) is all you need: On the influence of random seeds in deep learning architectures for computer vision," 2021, *arXiv:2109.08203*.

[55] D. Masters and C. Luschi, "Revisiting small batch training for deep neural networks," 2018, *arXiv:1804.07612*.

[56] M. K. Pal and A. Porwal, "Destriping of hyperion images using low-pass-filter and local-brightness-normalization," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Milan, Italy, Jul. 2015, pp. 3509–3512.

[57] *Hyperion Bhilwara Hyperspectral Data Cube*. Accessed: Aug. 2, 2020. [Online]. Available: https://earthexplorer.usgs.gov/

[58] N. Puletti, N. Camarretta, and P. Corona, "Evaluating EO1-hyperion capability for mapping conifer and broadleaved forests," *Eur. J. Remote Sens.*, vol. 49, no. 1, pp. 157–169, Feb. 2017.

[59] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.

[60] C.-H. Lin and T.-H. Lin, "All-addition hyperspectral compressed sensing for metasurface-driven miniaturized satellite," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503215.

[61] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[62] C.-H. Lin, C.-Y. Chi, L. Chen, D. J. Miller, and Y. Wang, "Detection of sources in non-negative blind source separation by minimum description length criterion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 9, pp. 4022–4037, Sep. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.

[63] K. Xie, G. Zhou, J. Yang, Z. He, and S. Xie, "Eliminating the permutation ambiguity of convolutive blind source separation by using coupled frequency bins," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 2, pp. 589–599, Feb. 2020.

[64] C. Zhou and M. R. D. Rodrigues, "Hyperspectral blind unmixing using a double deep image prior," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 16478–16492, Nov. 2024.

[65] H. Yu, B. Kong, Y. Hou, X. Xu, T. Chen, and X. Liu, "A critical review on applications of hyperspectral remote sensing in crop monitoring," *Experim. Agricult.*, vol. 58, p. 26, Jul. 2022.

[66] C. Weber et al., "Hyperspectral imagery for environmental urban planning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 1628–1631.

[67] C.-H. Lin, C.-Y. Chi, Y.-H. Wang, and T.-H. Chan, "A fast hyperplane-based minimum-volume enclosing simplex algorithm for blind hyperspectral unmixing," *IEEE Trans. Signal Process.*, vol. 64, no. 8, pp. 1946–1961, Apr. 2016.

[68] E. Pan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "SQAD: Spatial–spectral quasi-attention recurrent network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524814.

[69] M. Li, J. Liu, Y. Fu, Y. Zhang, and D. Dou, "Spectral enhanced rectangle transformer for hyperspectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Apr. 2023, pp. 5805–5814.

[70] A. Dixit, A. K. Gupta, P. Gupta, S. Srivastava, and A. Garg, "UNFOLD: 3-D U-Net, 3-D CNN, and 3-D transformer-based hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5529710.

[71] Y. Wu, J. Zhang, and D. Liu, "Predictive filtering integrated generative remote sensing hyperspectral image inpainting," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.



Chia-Hsiang Lin (Senior Member, IEEE) received the B.S. degree in electrical engineering and the Ph.D. degree in communications engineering from National Tsing Hua University (NTHU), Hsinchu, Taiwan, in 2010 and 2016, respectively.

From 2015 to 2016, he was a Visiting Student with Virginia Tech, Arlington, VA, USA. He is currently an Associate Professor with the Department of Electrical Engineering and the Miin Wu School of Computing, National Cheng Kung University (NCKU), Tainan, Taiwan. Before joining NCKU, he held a research positions with The Chinese University of Hong Kong, Hong Kong (2014 and 2017), NTHU from 2016 to 2017, and the University of Lisbon (ULisboa), Lisbon, Portugal, from 2017 to 2018. He was an Assistant Professor with the Center for Space and Remote Sensing Research, National Central University, Taoyuan, Taiwan, in 2018, and a Visiting Professor with ULisboa, in 2019. His research interests include network science, quantum computing, convex geometry and optimization, blind signal processing, and imaging science.

Dr. Lin received the Emerging Young Scholar Award (The 2030 Cross-Generation Program) by the National Science and Technology Council (NSTC) from 2023 to 2027, the Future Technology Award from NSTC, in 2022, the Outstanding Youth Electrical Engineer Award from Chinese Institute of Electrical Engineering (CIEE), in 2022, the Best Young Professional Member Award from the IEEE Tainan Section, in 2021, the Prize Paper Award from the IEEE Geoscience and Remote Sensing Society (GRS-S), in 2020, the Top Performance Award from the Social Media Prediction Challenge at ACM Multimedia, in 2020, and the 3rd Place from AIM Real World Super-Resolution Challenge at the IEEE International Conference on Computer Vision (ICCV), in 2019. He received the Ministry of Science and Technology (MOST) Young Scholar Fellowship, together with the EINSTEIN Grant Award, from 2018 to 2023. In 2016, he was a recipient of the Outstanding Doctoral Dissertation Award from Chinese Image Processing and Pattern Recognition Society and the Best Doctoral Dissertation Award from IEEE GRS-S.



Si-Sheng Young (Student Member, IEEE) is currently pursuing the Ph.D. degree with the Intelligent Hyperspectral Computing Laboratory (IHCL), Department of Electrical Engineering, National Cheng Kung University (NCKU), Tainan, Taiwan.

His research interests include convex optimization, deep learning, anomaly detection, data fusion, and imaging inverse problems.

Mr. Young received the Merit Award from the Grand Challenge "Computing for the Future," Miin Wu School of Computing, NCKU, as well as the highly competitive "Pan Wen Yuan Scholarship" from the Industrial Technology Research Institute, Hsinchu, Taiwan, in 2023. In 2024, he received a highly competitive "Scholarship Pilot Program to Cultivate Outstanding Doctoral Students" from the National Science and Technology Council (NSTC), Taipei, Taiwan.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:56:46 UTC from IEEE Xplore. Restrictions apply.